

DATA 260 HOMEWORK 2

Grocery Supply and Recall Notices Stateful Agent Extension

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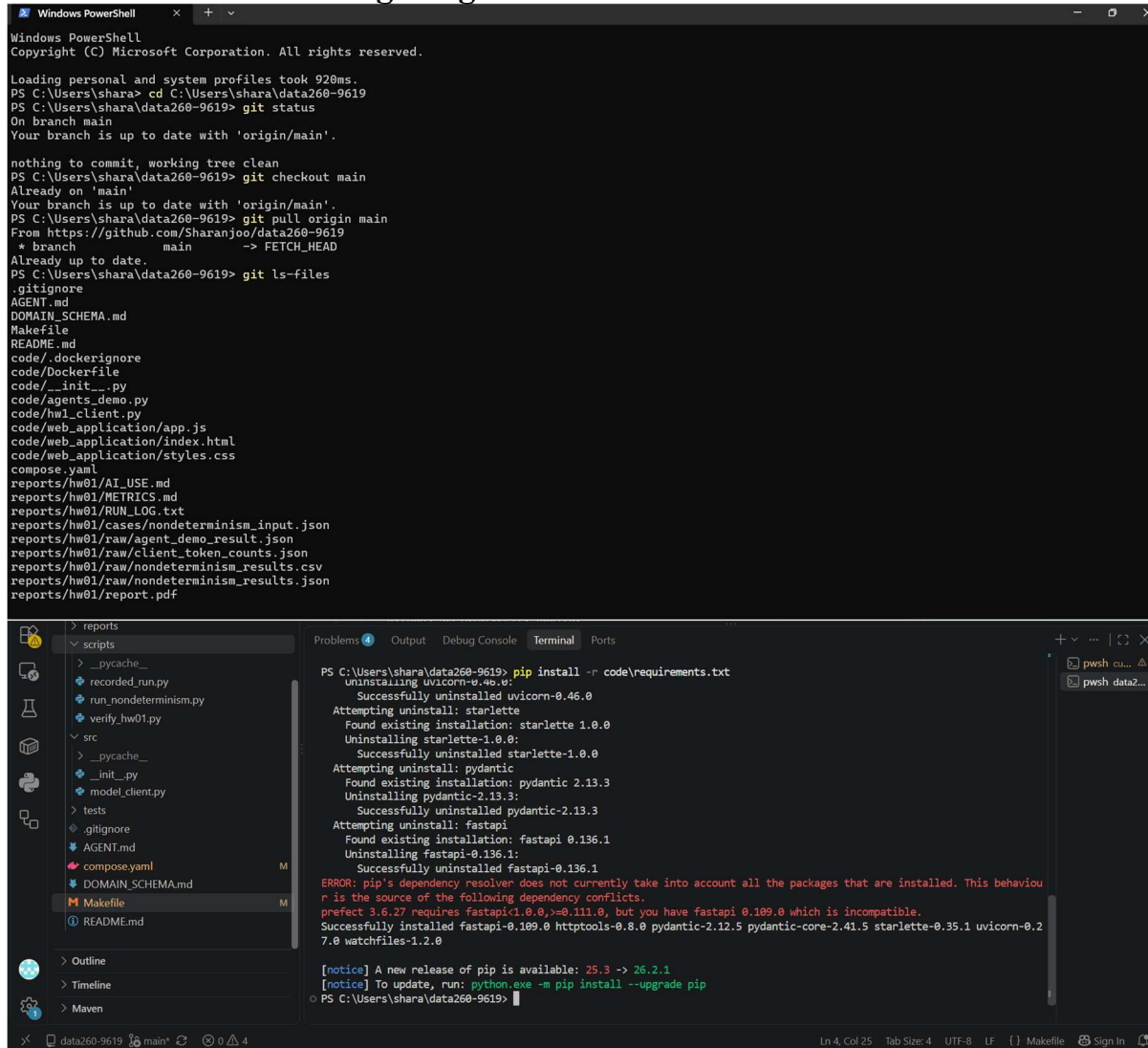
Section 0 | Personal Configuration and Domain

Value	Result
SID4	9619
PORT_BASE	8619
PREFIX	s9619
SEED	9619
VERIFY_SEED	269619
DOMAIN_ID	3
Assigned domain	Grocery supply and recall notices
Local model	qwen3:8b

GitHub repository: <https://github.com/Sharanjoo/data260-9619>

Implementation commit reference: b80ec13b681f049bc3fd8ba8edb69d850dc5f217

Pre-check before starting assignment 2:



The screenshot shows a Windows PowerShell terminal window and a VS Code editor interface. The PowerShell window displays the output of several git commands and a file listing. The VS Code editor shows a file explorer on the left with a project structure, a terminal window in the center, and a status bar at the bottom.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Loading personal and system profiles took 920ms.
PS C:\Users\shara> cd C:\Users\shara\data260-9619
PS C:\Users\shara\data260-9619> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\shara\data260-9619> git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\shara\data260-9619> git pull origin main
From https://github.com/Sharanjoo/data260-9619
* branch      main      -> FETCH_HEAD
Already up to date.
PS C:\Users\shara\data260-9619> git ls-files
.gitignore
AGENT.md
DOMAIN_SCHEMA.md
Makefile
README.md
code/.dockerignore
code/Dockerfile
code/_init_.py
code/agents_demo.py
code/hw1_client.py
code/web_application/app.js
code/web_application/index.html
code/web_application/styles.css
compose.yaml
reports/hw01/AI_USE.md
reports/hw01/METRICS.md
reports/hw01/RUN_LOG.txt
reports/hw01/cases/nondeterminism_input.json
reports/hw01/raw/agent_demo_result.json
reports/hw01/raw/client_token_counts.json
reports/hw01/raw/nondeterminism_results.csv
reports/hw01/raw/nondeterminism_results.json
reports/hw01/report.pdf
```

The VS Code interface shows a file explorer on the left with a project structure. The terminal window in the center displays the output of the command `pip install -r code/requirements.txt`. The status bar at the bottom shows the current file is `data260-9619` on the `main` branch, with 4 changes.

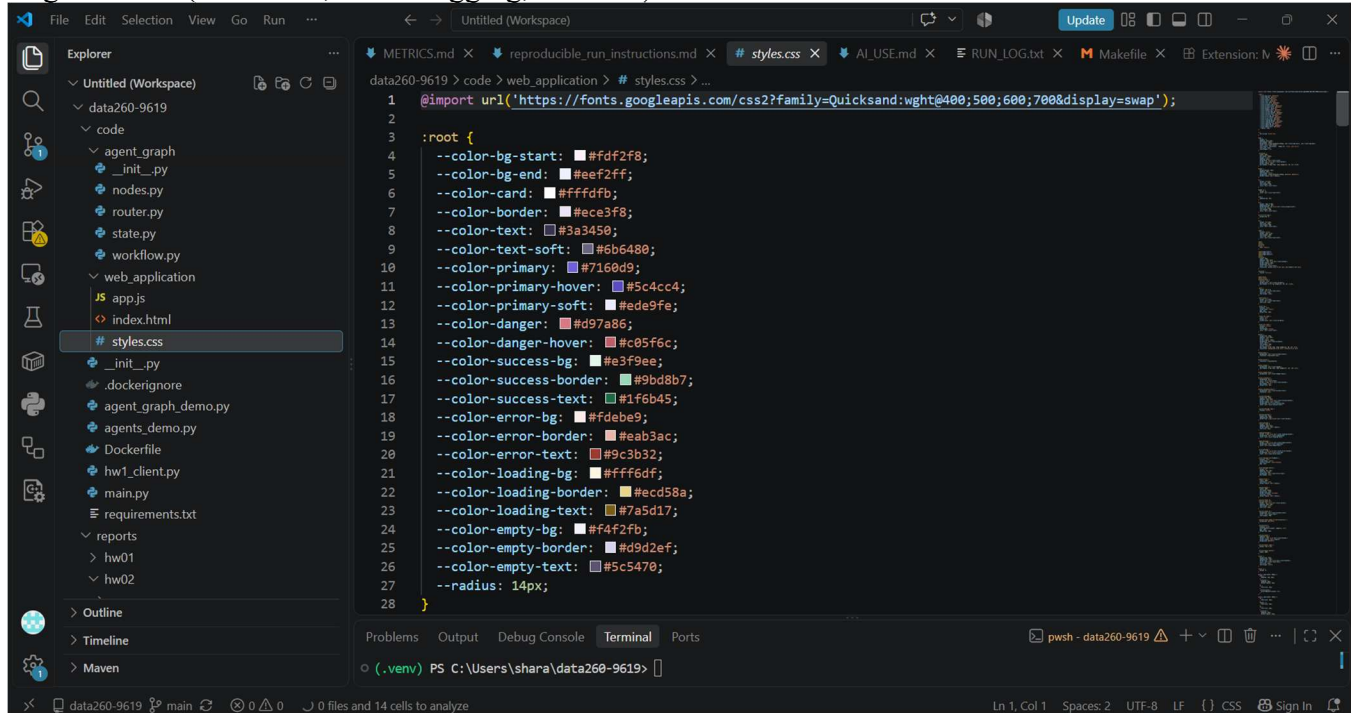
```
PS C:\Users\shara\data260-9619> pip install -r code/requirements.txt
uninstalling uvicorn-0.46.0:
Successfully uninstalled uvicorn-0.46.0
Attempting uninstall: starlette
Found existing installation: starlette 1.0.0
Uninstalling starlette-1.0.0:
Successfully uninstalled starlette-1.0.0
Attempting uninstall: pydantic
Found existing installation: pydantic 2.13.3
Uninstalling pydantic-2.13.3:
Successfully uninstalled pydantic-2.13.3
Attempting uninstall: fastapi
Found existing installation: fastapi 0.136.1
Uninstalling fastapi-0.136.1:
Successfully uninstalled fastapi-0.136.1
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour
is the source of the following dependency conflicts.
prefect 3.6.27 requires fastapi<1.0.0,>=0.111.0, but you have fastapi 0.109.0 which is incompatible.
Successfully installed fastapi-0.109.0 httpx-0.27.0 pydantic-2.12.5 pydantic-core-2.41.5 starlette-0.35.1 uvicorn-0.2
7.0 watchfiles-1.2.0

[notice] A new release of pip is available: 25.3 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\shara\data260-9619>
```

The status bar at the bottom of VS Code shows: `data260-9619` `main` `4` changes. The bottom right corner shows: `Ln 4, Col 25` `Tab Size: 4` `UTF-8` `LF` `Makefile` `Sign In`.

PART I | Web Application

For this homework I kept the same Grocery Supply and Recall Notices form from HW1 with all the same fields. I added a new Notices section that lists every recall notice, with a loading state, an empty state, and an error state with a retry button. I also restyled the whole page with a pastel colour theme using CSS custom properties, so it looks less plain than the HW1 version. Everything else about the original form (validation, JSON logging, closures) works the same as before.



```
PS C:\Users\shara\data260-9619> git add code\web_application\index.html code\web_application\styles.css
warning: in the working copy of 'code/web_application/index.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'code/web_application/styles.css', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\shara\data260-9619> git commit -m "hw2 part1: add responsive notices list markup and loading/empty/error state styles"
[main 5ce40b8] hw2 part1: add responsive notices list markup and loading/empty/error state styles
2 files changed, 313 insertions(+), 47 deletions(-)
PS C:\Users\shara\data260-9619> git push origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 3.05 KiB | 520.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/Sharanjoo/data260-9619.git
b5d1d70..5ce40b8 main -> main
PS C:\Users\shara\data260-9619> git add code\web_application\styles.css
PS C:\Users\shara\data260-9619> git commit -m "hw2 part1: pastel visual redesign of the notices page"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\shara\data260-9619> git push origin main
Everything up-to-date
PS C:\Users\shara\data260-9619>
```

New look of my web page

← → ↻ 🏠 ⓘ localhost:8619 🔍 ☆ 📄 🌐 ⋮

Grocery Supply and Recall Notices

Use this form to submit a grocery supply or product recall notice.

Submit a Notice

Product name *

High Protein Yogurt

Brand or supplier *

Chobani

Submitter email *

sharanjoseph16@gmail.com

Notice category *

Quality Withdrawal

Notice description *

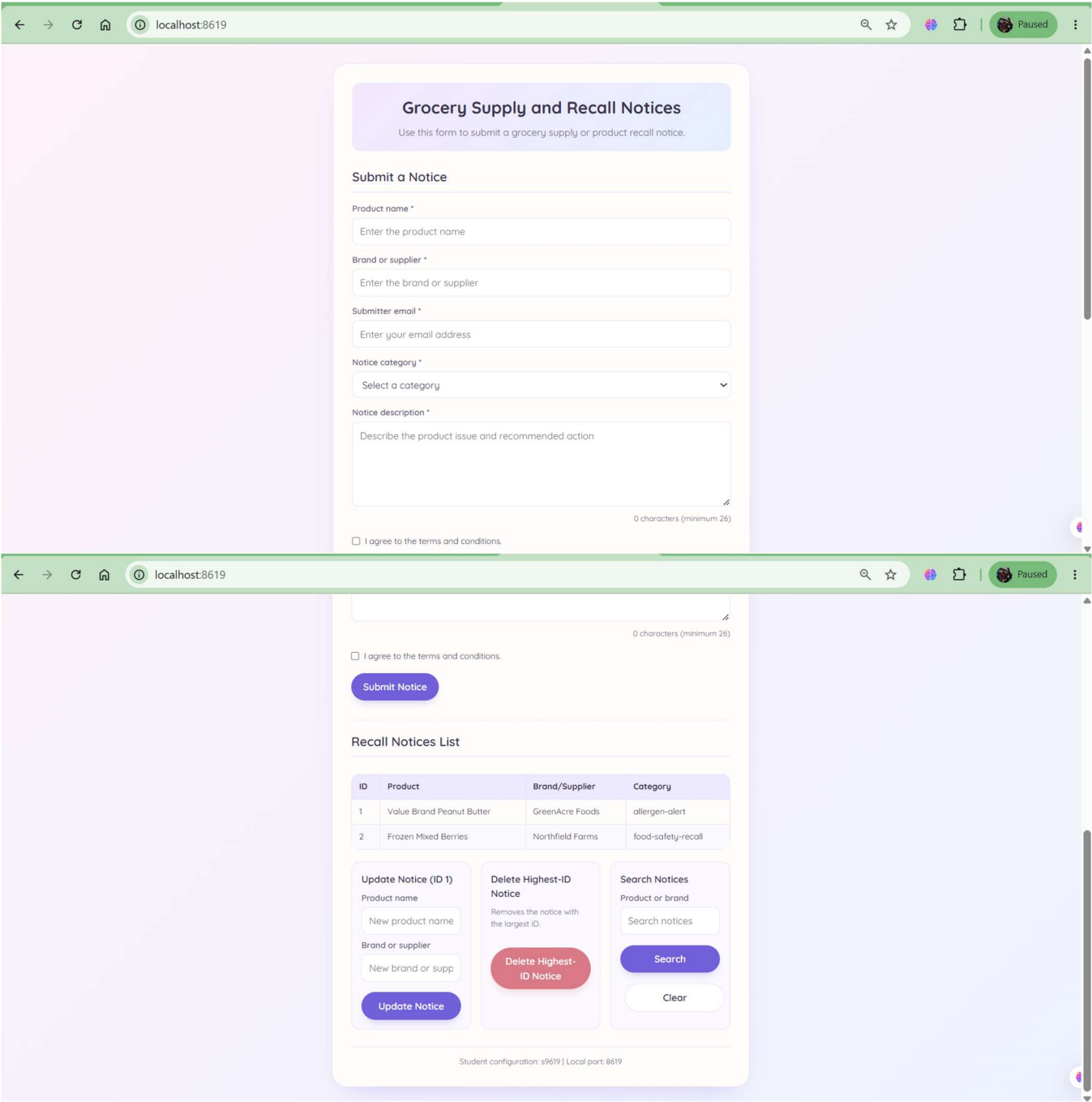
Not the expected consistency

29 characters (minimum 26)

☒ I agree to the terms and conditions.

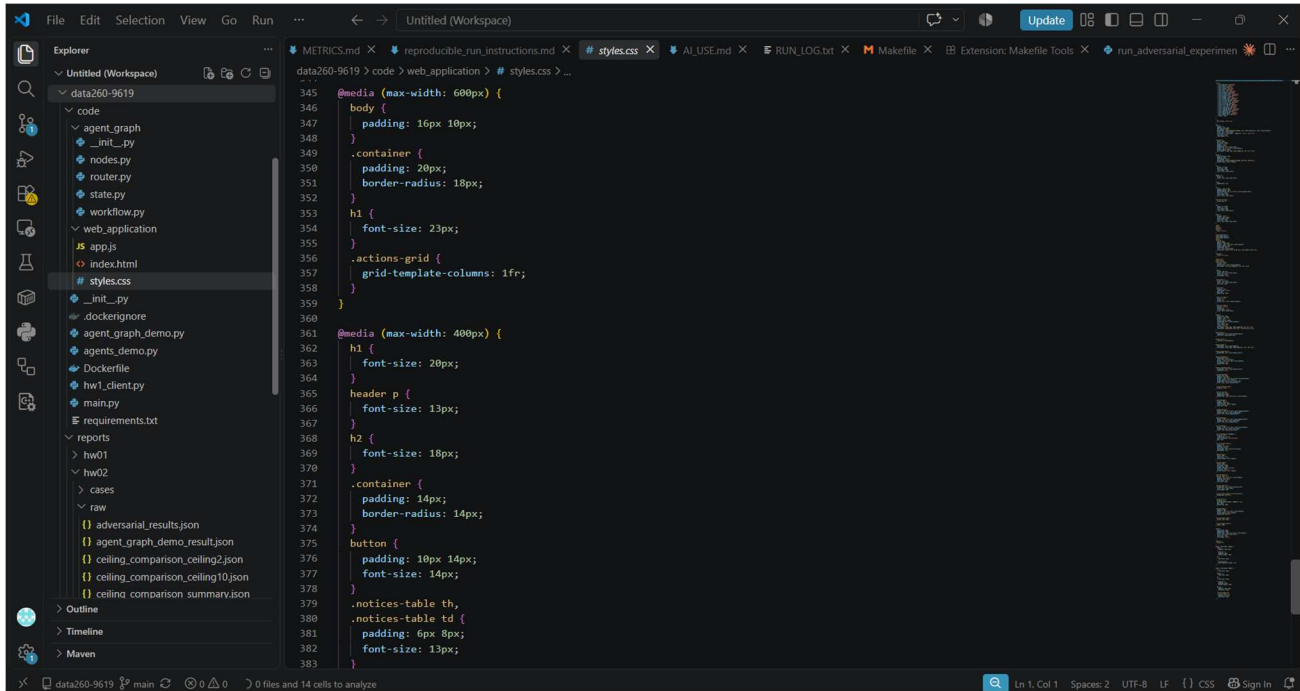
Submit Notice

Desktop view: The screenshot below shows the full page on a normal desktop width, with the form on top and the notices table below it.



Narrow / mobile view (375px)

The screenshot below shows the same page at a 375px width, to confirm the layout is still usable on a small screen. Z



Grocery Supply and Recall Notices

Use this form to submit a grocery supply or product recall notice.

Submit a Notice

Product name *

Enter the product name

Brand or supplier *

Enter the brand or supplier

Submitter email *

Enter your email address

Notice category *

<DOCTYPE html>

<html lang="en">

</html>

body

Filter

element.style {

}

@media (max-width: 600px) {

body {

padding: 16px 10px;

}

}

body {

margin: 0;

padding: 32px 16px;

min-height: 100vh;

background: linear-gradient(135deg, var(--color-bg-start), var(--color-bg-end));

color: var(--color-text);

font-family: 'Quicksand', 'Segoe UI', Arial, sans-serif;

font-size: 16px;

line-height: 1.5;

}

Console

AI assistance

top

Filter

Default levels

No issues

Notice description *

Describe the product issue and recommended action

0 characters (minimum 26)

☐ I agree to the terms and conditions.

Submit Notice

Recall Notices List

ID	Product	Brand/Supplier	Category
1	Value Brand Peanut Butter	GreenAcre Foods	all
2	Frozen Mixed Berries	Northfield Farms	fo re

<DOCTYPE html>

<html lang="en">

</html>

body

Filter

element.style {

}

@media (max-width: 600px) {

body {

padding: 16px 10px;

}

}

body {

margin: 0;

padding: 32px 16px;

min-height: 100vh;

background: linear-gradient(135deg, var(--color-bg-start), var(--color-bg-end));

color: var(--color-text);

font-family: 'Quicksand', 'Segoe UI', Arial, sans-serif;

font-size: 16px;

line-height: 1.5;

}

Console

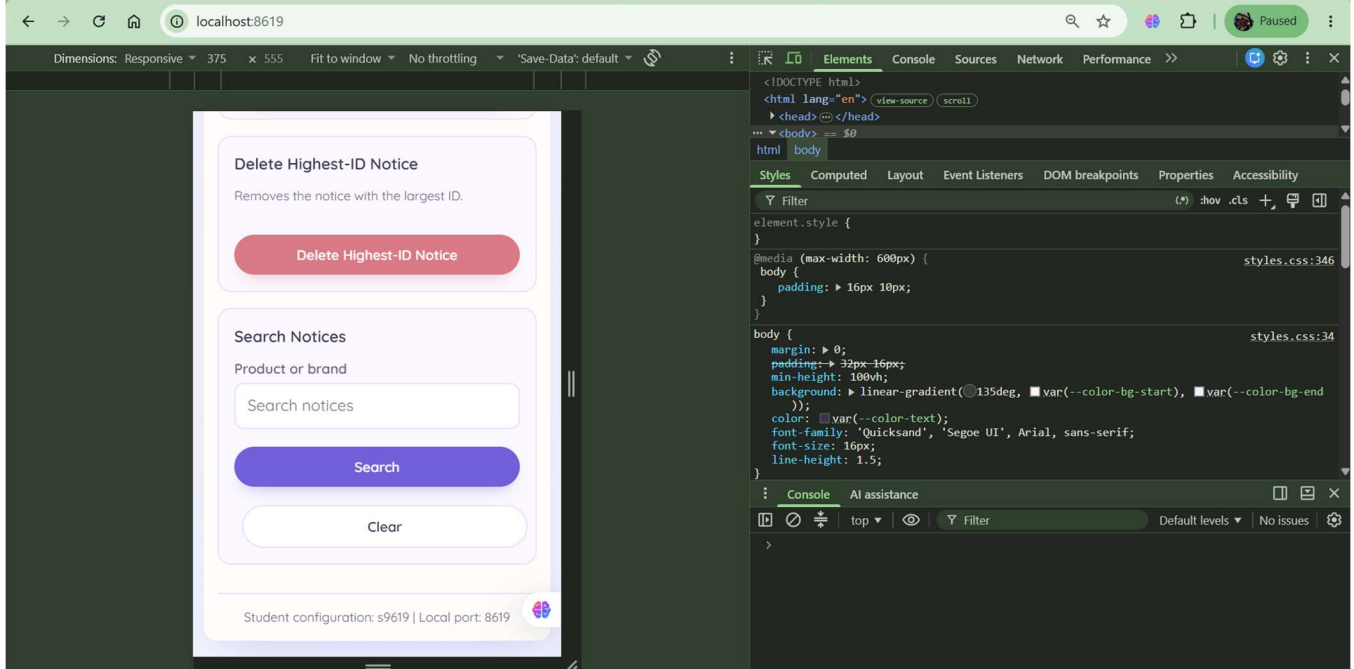
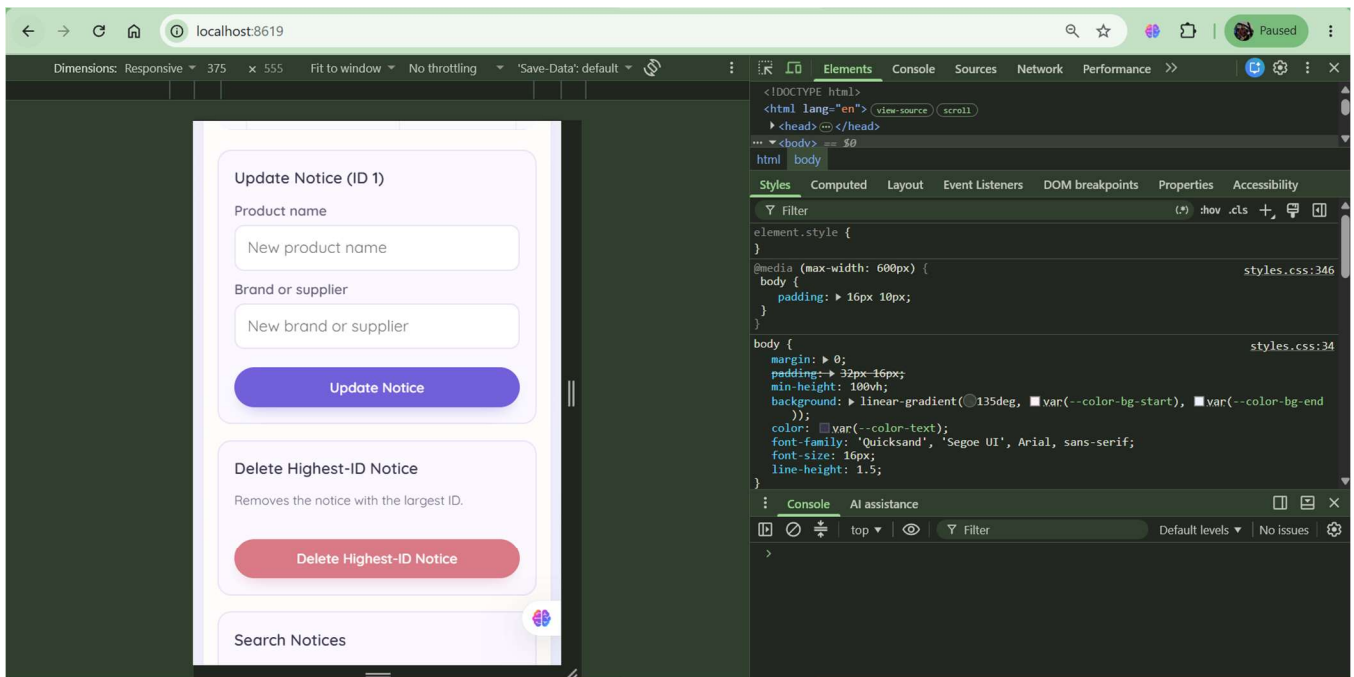
AI assistance

top

Filter

Default levels

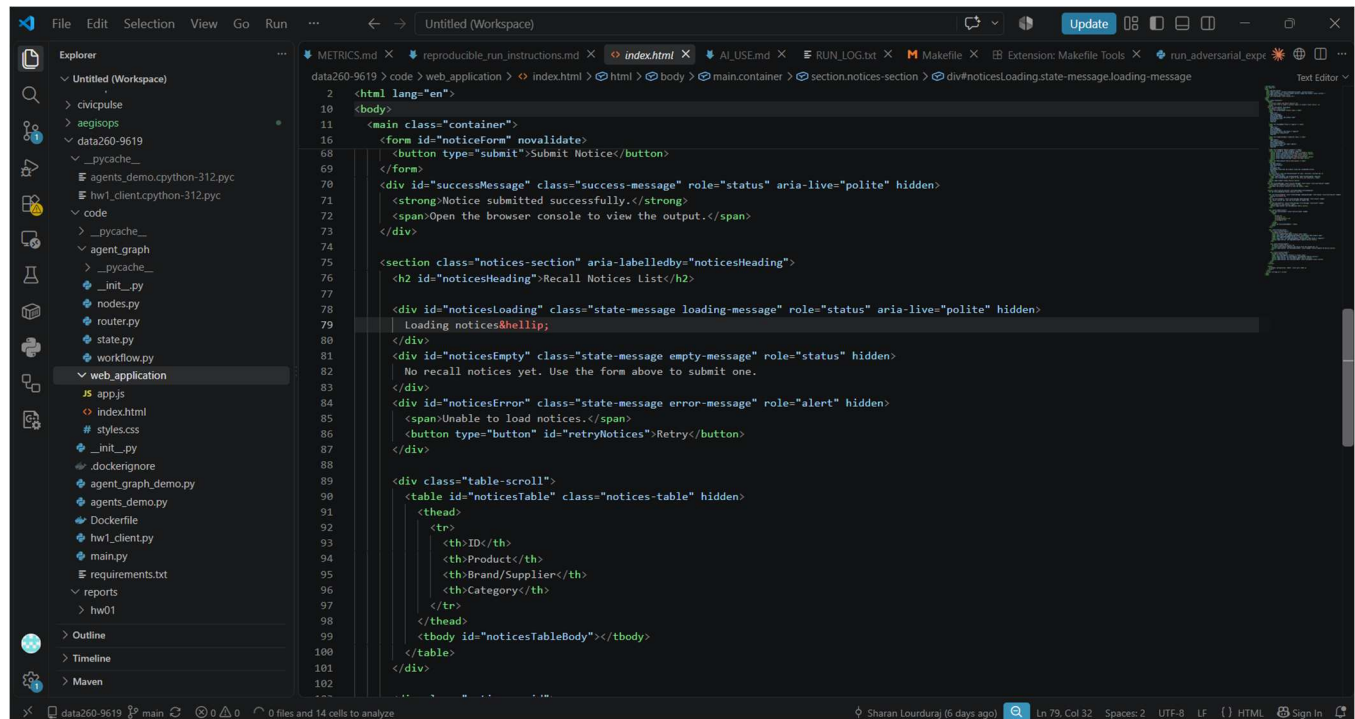
No issues



Loading, empty, and error states

The notices table shows a loading message while it is fetching data, an empty-state message if there are no notices, and an error message with a Retry button if the request fails.

Index.html

A screenshot of the Visual Studio Code editor interface. The Explorer sidebar on the left shows a project structure with folders like 'data260-9619' and 'web_application'. The main editor area displays the 'index.html' file. The code is an HTML document with a form for submitting notices and a table for displaying them. It includes conditional rendering logic for loading, empty, and error states using ARIA roles and hidden attributes. The status bar at the bottom indicates the file is at line 79, column 32, with 2 spaces and UTF-8 encoding.

```
data260-9619 > code > web_application > index.html > html > body > main.container > section.notices-section > div#noticesLoading.state-message.loading-message
2  <html lang="en">
10 <body>
11   <main class="container">
16     <form id="noticeForm" novalidate>
68       <button type="submit">Submit Notice</button>
69     </form>
70     <div id="successMessage" class="success-message" role="status" aria-live="polite" hidden>
71       <strong>Notice submitted successfully.</strong>
72       <span>Open the browser console to view the output.</span>
73     </div>
74     <section class="notices-section" aria-labelledby="noticesHeading">
75       <h2 id="noticesHeading">Recall Notices List</h2>
76       <div id="noticesLoading" class="state-message loading-message" role="status" aria-live="polite" hidden>
77         Loading notices&hellip;
78       </div>
79       <div id="noticesEmpty" class="state-message empty-message" role="status" hidden>
80         No recall notices yet. Use the form above to submit one.
81       </div>
82       <div id="noticesError" class="state-message error-message" role="alert" hidden>
83         <span>Unable to load notices.</span>
84         <button type="button" id="retryNotices">Retry</button>
85       </div>
86       <div class="table-scroll">
87         <table id="noticesTable" class="notices-table" hidden>
88           <thead>
89             <tr>
90               <th>ID</th>
91               <th>Product</th>
92               <th>Brand/Supplier</th>
93               <th>Category</th>
94             </tr>
95           </thead>
96           <tbody id="noticesTableBody"></tbody>
97         </table>
98       </div>
99     </div>
100   </main>
101 </body>
102 </html>
```

App.js

A screenshot of the Visual Studio Code editor interface showing the 'App.js' file. The code defines a 'showNoticesState' function that updates the 'hidden' attribute of various DOM elements based on the current state ('loading', 'empty', 'error', 'data'). The Explorer sidebar on the left shows the project structure, and the status bar at the bottom indicates the file is at line 60, column 10.

```
state.py
workflow.py
web_application
JS app.js
index.html
styles.css
_init_.py
.dockernignore
agent_graph_demo.py
agents_demo.py
Dockerfile
hw1_client.py
main.py
requirements.txt
reports
hw01

state.py
workflow.py
web_application
JS app.js
index.html
styles.css
_init_.py
.dockernignore
agent_graph_demo.py

56   });
57   });
58   // ---- Notices list: loading / empty / error / data states ----
59
60   function showNoticesState(state) {
61     noticesLoading.hidden = state !== "loading";
62     noticesEmpty.hidden = state !== "empty";
63     noticesError.hidden = state !== "error";
64     noticesTable.hidden = state !== "data";
65   }
66
```

Loading state

localhost:8619

Dimensions: Responsive 500 x 740 75% Slow 4G Save-Data: default

Describe the product issue and recommended action

0 characters (minimum 26)

☐ I agree to the terms and conditions.

Submit Notice

Recall Notices List

Loading notices...

Update Notice (ID 1)

Product name

New product name

Brand or supplier

New brand or supplier

Update Notice

Network

Filter

Name	Status	Type	Initiator	Size	Time
content-all.css	200	fetch	content-all.js:514	554 kB	8 ms
fonts.css	200	fetch	content-all.js:514	6.7 kB	6 ms
enjson	200	fetch	content-all.js:518	232 kB	7 ms
notices	(pending)	fetch	app.js:88	0.0 kB	Pending

4 / 11 requests 792 kB / 836 kB transferred 793 kB / 889 kB resources Finish: 3.18 s DOMContentLoaded: 1.12 s Load: 2.1 s

Console AI assistance Request conditions X

Enable blocking and throttling +

Nothing throttled or blocked

To throttle or block a network request, add a rule here manually or via the network panel's context menu. [Learn more](#)

Add rule

Empty state

localhost:8619

Dimensions: Responsive 500 x 740 75% Slow 4G Save-Data: default

Describe the product issue and recommended action

0 characters (minimum 26)

☐ I agree to the terms and conditions.

Submit Notice

Recall Notices List

No recall notices yet. Use the form above to submit one.

Update Notice (ID 1)

Product name

New product name

Brand or supplier

New brand or supp

Update Notice

Delete Highest-ID Notice

Removes the notice with the largest ID.

Delete Highest-ID Notice

Search Notices

Product or brand

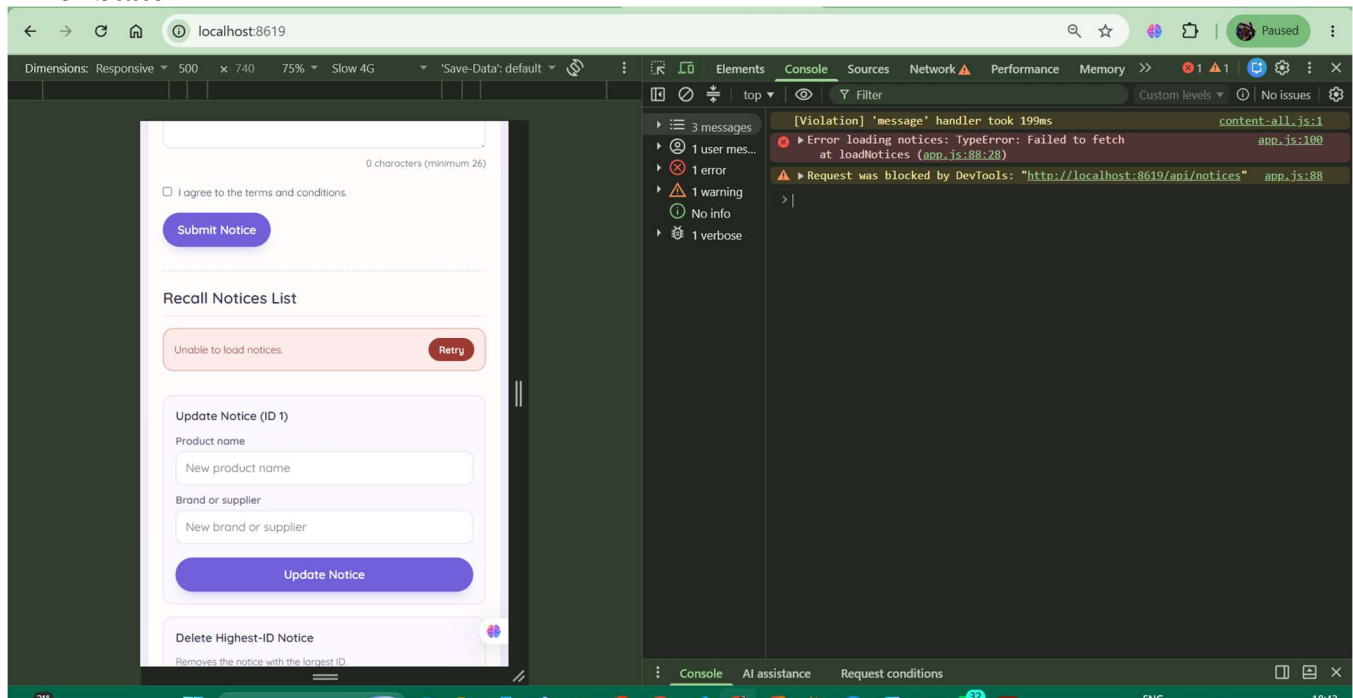
does

Search

Clear

Student configuration: s9619 | Local port: 8619

Error State



PART II | FastAPI Backend and Docker

For Part II I replaced the static nginx setup from HW1 with a real backend. I wrote code/main.py using Fast API, which now serves both the web page and a JSON API from the same port (my PORT_BASE, 8619).

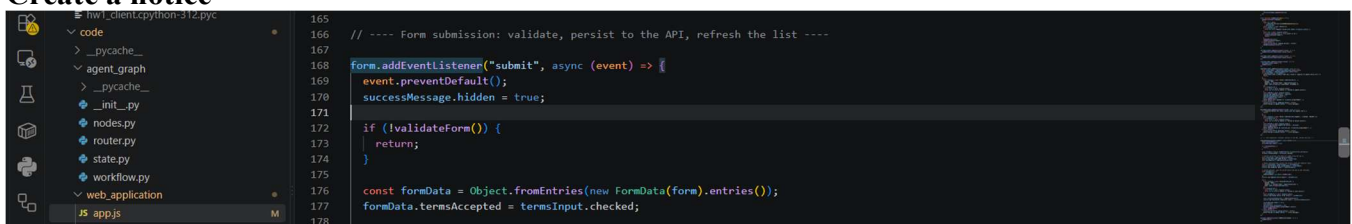
The API has these routes:

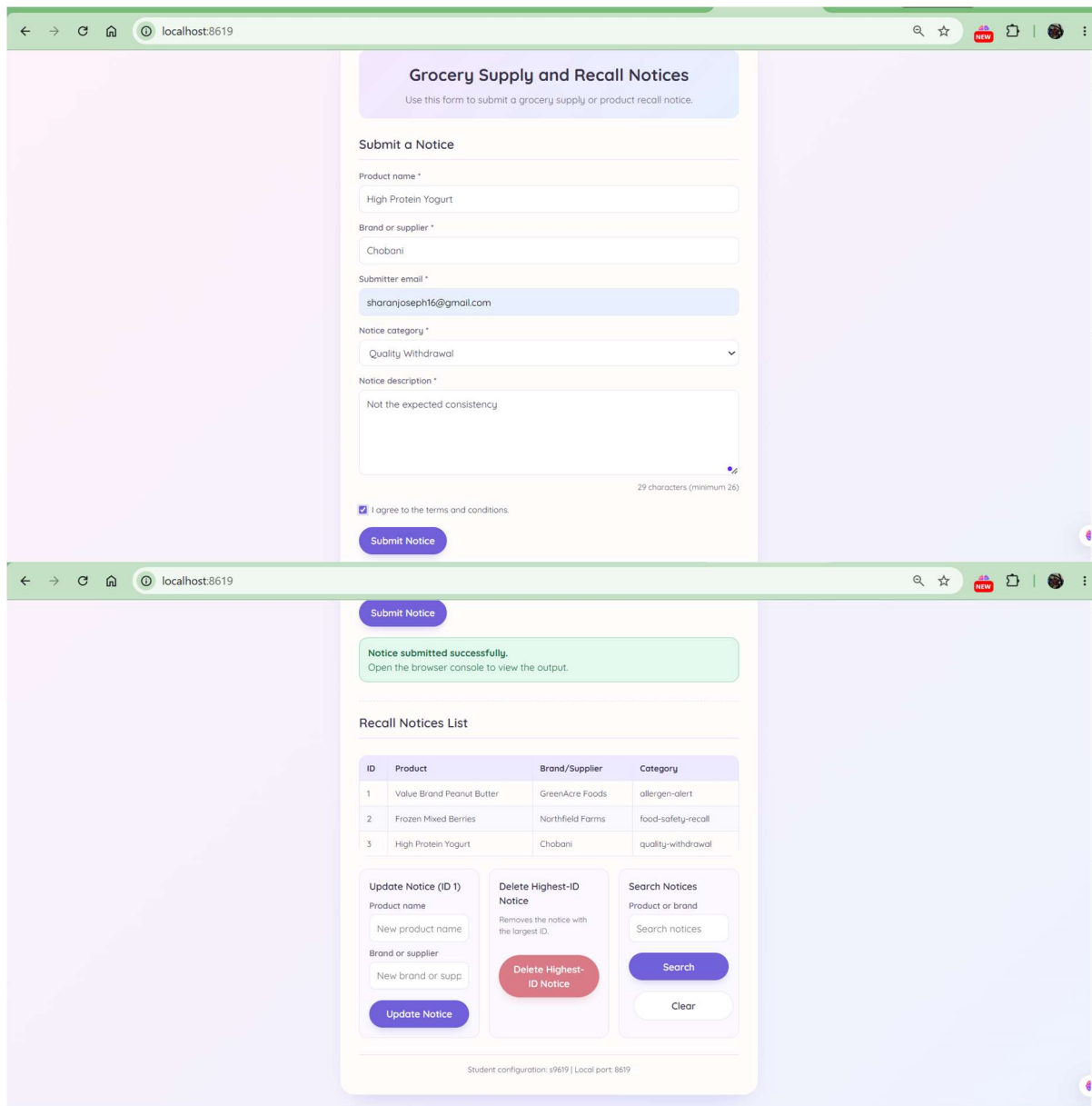
- GET /api/health a simple health check
- GET /api/notices returns the list of notices, and supports an optional search (q) parameter
- POST /api/notices creates a new notice from the form
- PUT /api/notices/{id} update the product name and brand name of a notice
- DELETE /api/notices/highest removes the notice with the highest ID

On the frontend, I connected the Update, Delete, and Search panels to these routes, so the page can create, update, delete, and search notices without a page reload.

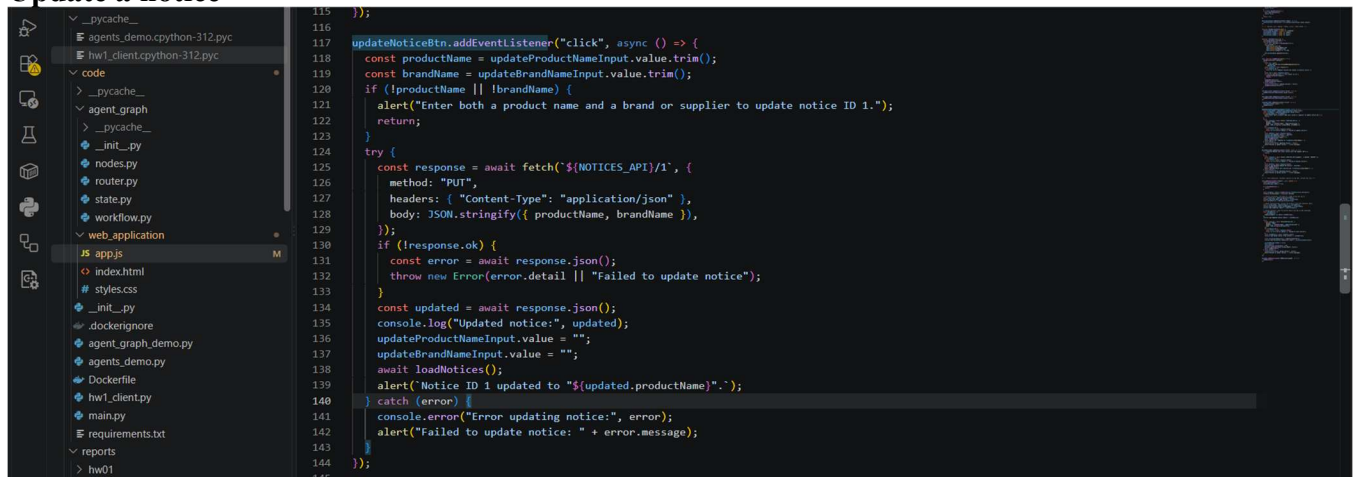
I also rebuilt the Docker image, so it runs this FastAPI app with uvicorn instead of nginx, since the app is no longer just static files.

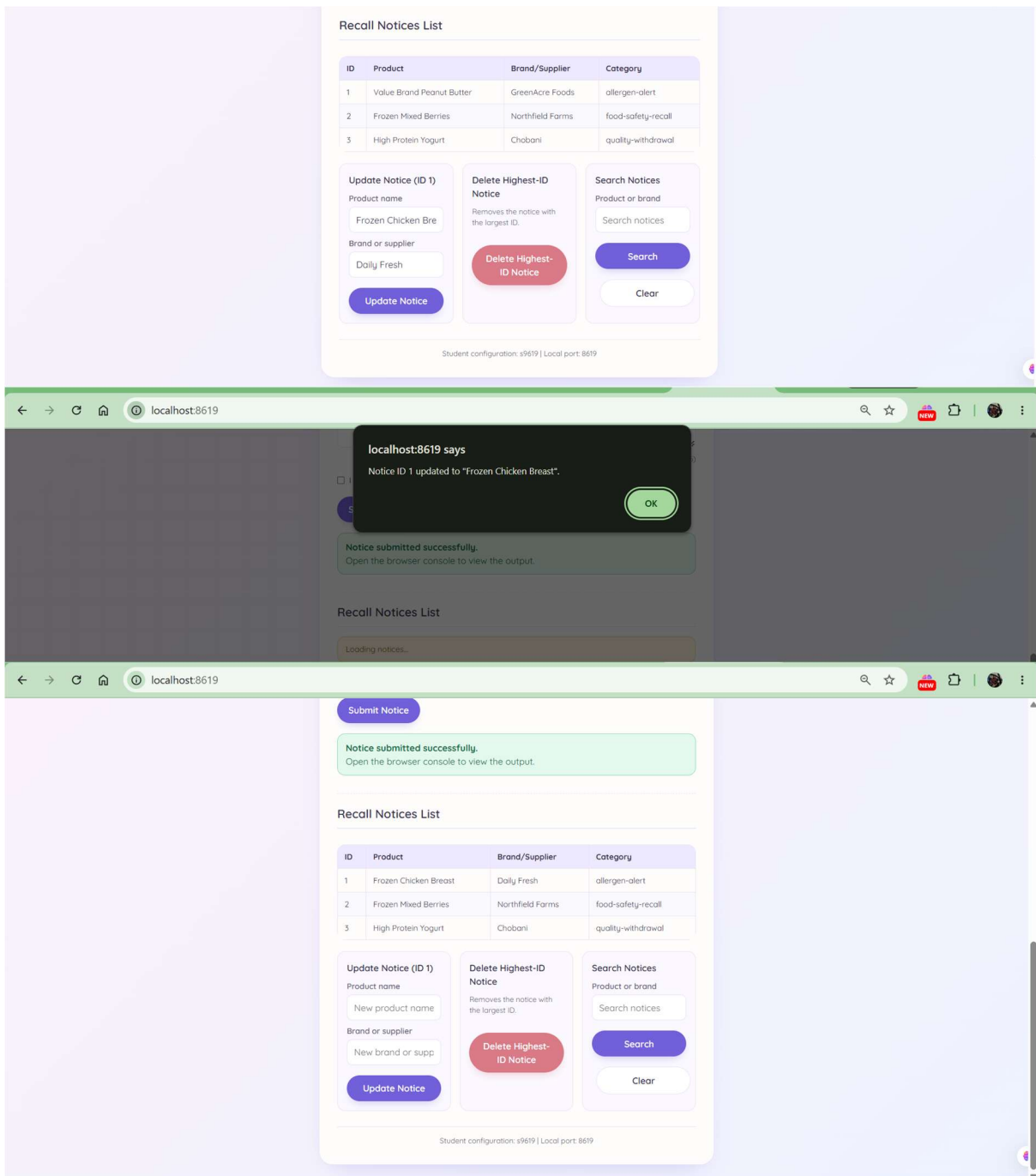
Create a notice





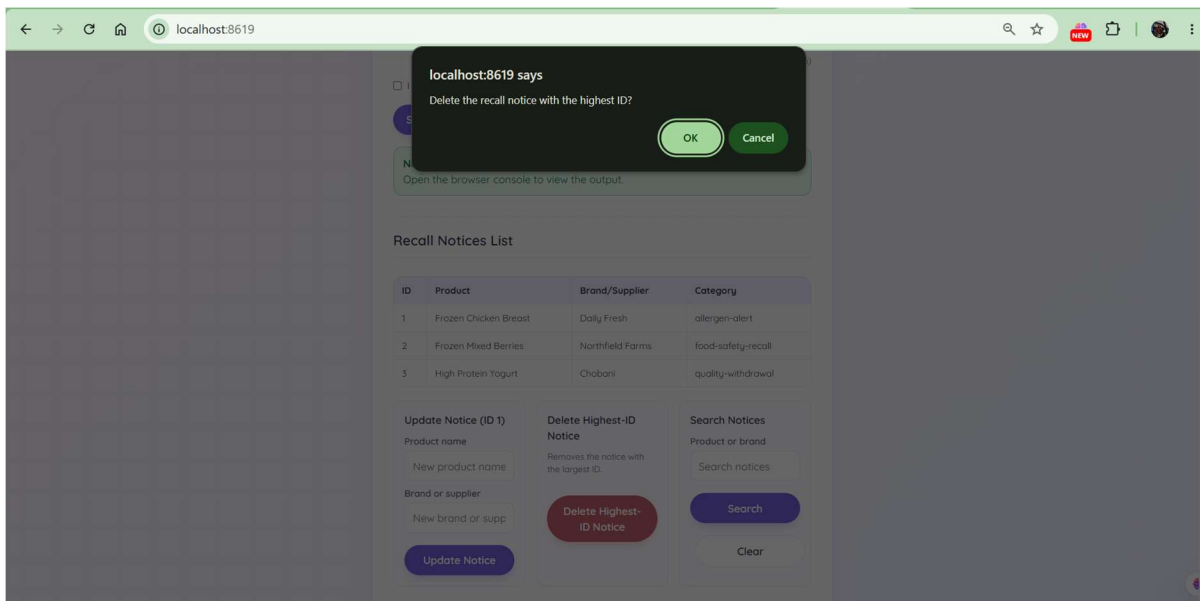
Update a notice



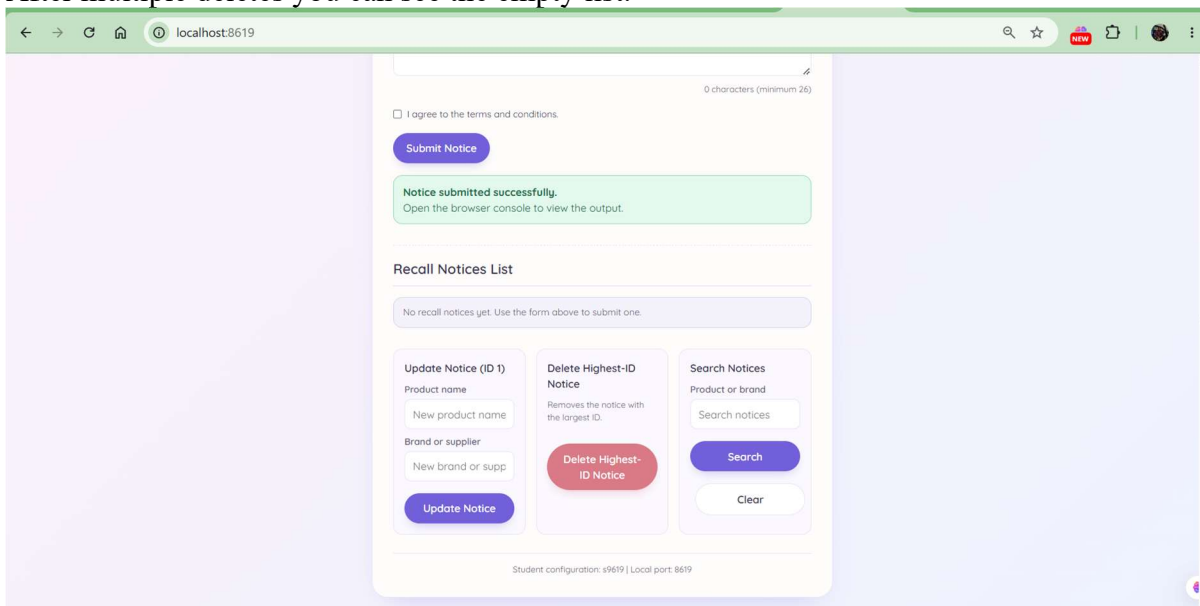


Delete a notice





After multiple deletes you can see the empty list.



Search notices

workflow.py

web_application

app.js

index.html

styles.css

106

107

108

109

110

111

});

searchNoticeBtn.addEventListener("click", () => {

loadNotices(searchQueryInput.value.trim());

});

localhost:8619

0 characters (minimum 26)

NEW

☐ I agree to the terms and conditions.

Submit Notice

Notice submitted successfully.

Open the browser console to view the output.

Recall Notices List

ID	Product	Brand/Supplier	Category
2	Frozen Mixed Berries	Northfield Farms	food-safety-recall

Update Notice (ID 1)

Product name

New product name

Brand or supplier

New brand or supp

Update Notice

Delete Highest-ID Notice

Removes the notice with the largest ID.

Delete Highest-ID Notice

Search Notices

Product or brand

Berries

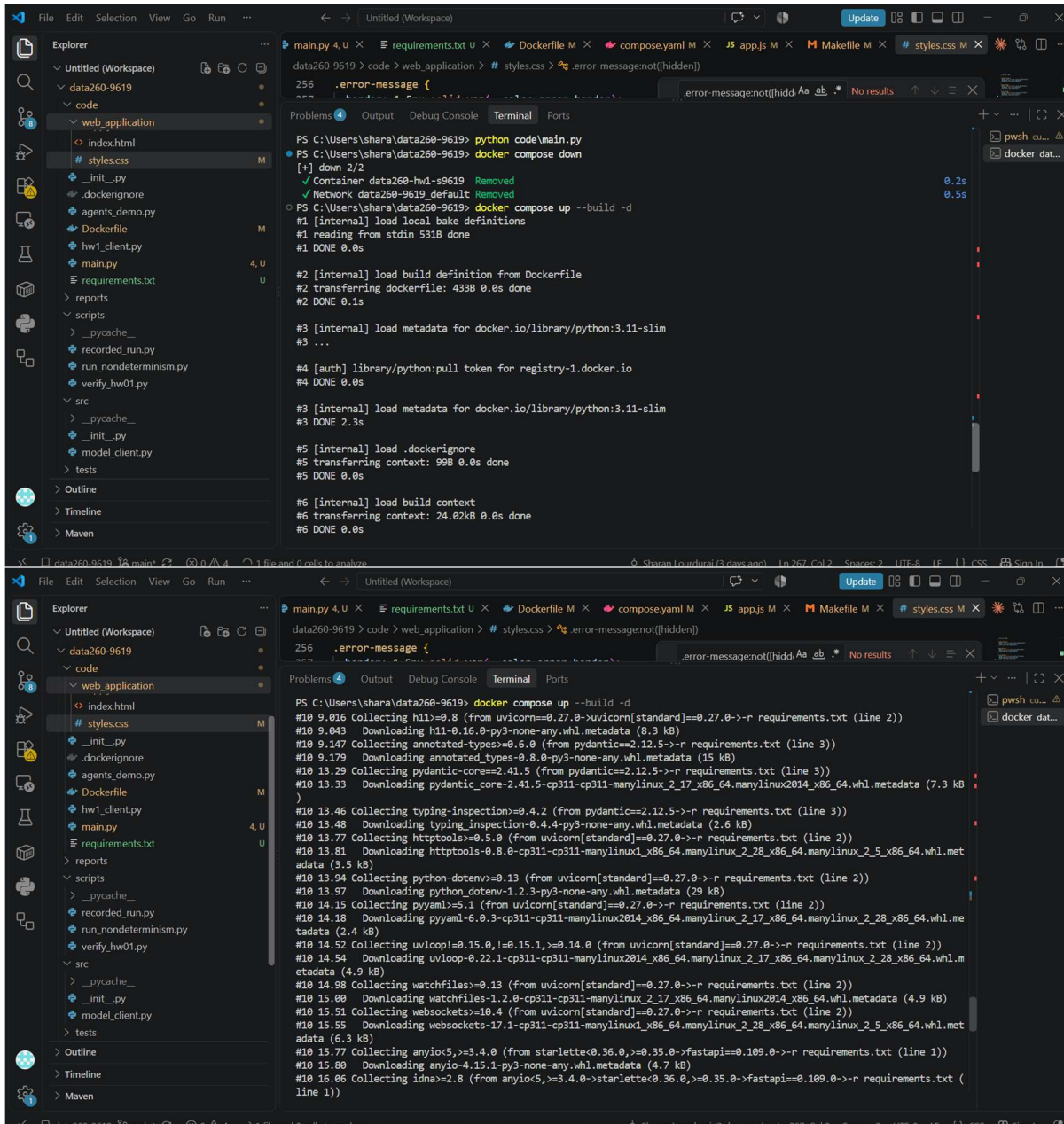
Search

Clear

Student configuration: s9619 | Local port: 8619

Docker deployment

I ran `docker compose up --build -d` and confirmed the container comes up healthy and the app is reachable at `http://localhost:8619`.



The top screenshot shows the VS Code interface with the Explorer view on the left, displaying the project structure. The main editor shows the `main.py` file. The terminal at the bottom displays the command `docker compose up --build -d` being executed, with progress bars for downloading Docker images.

```
PS C:\Users\shara\data260-9619> python code/main.py
data260-9619 > code > web_application > # styles.css > # error-messagenot([hidden])
256 .error-message {
    background-color: #f0f0f0;
    padding: 5px;
    border: 1px solid #ccc;
}

```

The bottom screenshot shows the VS Code interface with the Explorer view on the left, displaying the project structure. The main editor shows the `main.py` file. The terminal at the bottom displays the command `docker compose up --build -d` being executed, with progress bars for downloading Docker images.

```
PS C:\Users\shara\data260-9619> docker compose up --build -d
data260-9619 > code > web_application > # styles.css > # error-messagenot([hidden])
256 .error-message {
    background-color: #f0f0f0;
    padding: 5px;
    border: 1px solid #ccc;
}

```

```
data260-9619 > code > web_application > # styles.css > .error-message:not([hidden])
256 .error-message {
...
}

Problems Output Debug Console Terminal Ports
PS C:\Users\shara\data260-9619> docker compose up --build -d
#10 22.45 [notice] To update, run: pip install --upgrade pip
#10 DONE 23.7s

#11 [5/6] COPY main.py .
#11 DONE 0.2s

#12 [6/6] COPY web_application ./web_application
#12 DONE 0.1s

#13 exporting to image
#13 exporting layers
#13 exporting layers 6.2s done
#13 exporting manifest sha256:92b38446c8c2982039c37ac7eba91af5638a87f7284bec046cb07357c42254d4 0.1s done
#13 exporting config sha256:dc4aa2b8c1609b9e9ff2fa31e0105e48a4b510dc9a482081765e2d1f2a79d649 0.1s done
#13 exporting attestation manifest sha256:5ce5343f2f4067abc3808d12b8da8395dd31f8c960a43341124295e0a379b677 0.1s done
#13 exporting manifest list sha256:2de9926e357227f1cde0fc5004557abdf1025d0bb4bb97bd1e422b59407f92fd
#13 exporting manifest list sha256:2de9926e357227f1cde0fc5004557abdf1025d0bb4bb97bd1e422b59407f92fd 0.1s done
#13 naming to docker.io/library/data260-hw2-s9619:latest done
#13 unpacking to docker.io/library/data260-hw2-s9619:latest
#13 unpacking to docker.io/library/data260-hw2-s9619:latest 3.1s done
#13 DONE 9.8s

#14 resolving provenance for metadata file
#14 DONE 0.1s
[+] up 3/3
  Image data260-hw2-s9619:latest Built 45.6s
  Network data260-9619_default Created 0.2s
  Container data260-hw2-s9619 Started 2.6s
PS C:\Users\shara\data260-9619>
PS C:\Users\shara\data260-9619> curl http://localhost:8619/api/health
{"status": "ok", "port": 8619}
PS C:\Users\shara\data260-9619>
```

PART III | Stateful LangGraph Agent

For Part III I built a small stateful agent using LangGraph's StateGraph, in a new code/agent_graph/ folder. The agent takes a recall notice's title and content and produces exactly 3 tags and a summary of no more than 25 words, the same way the HW1 agent did, but this time with real state and a real control loop instead of a straight-line script.

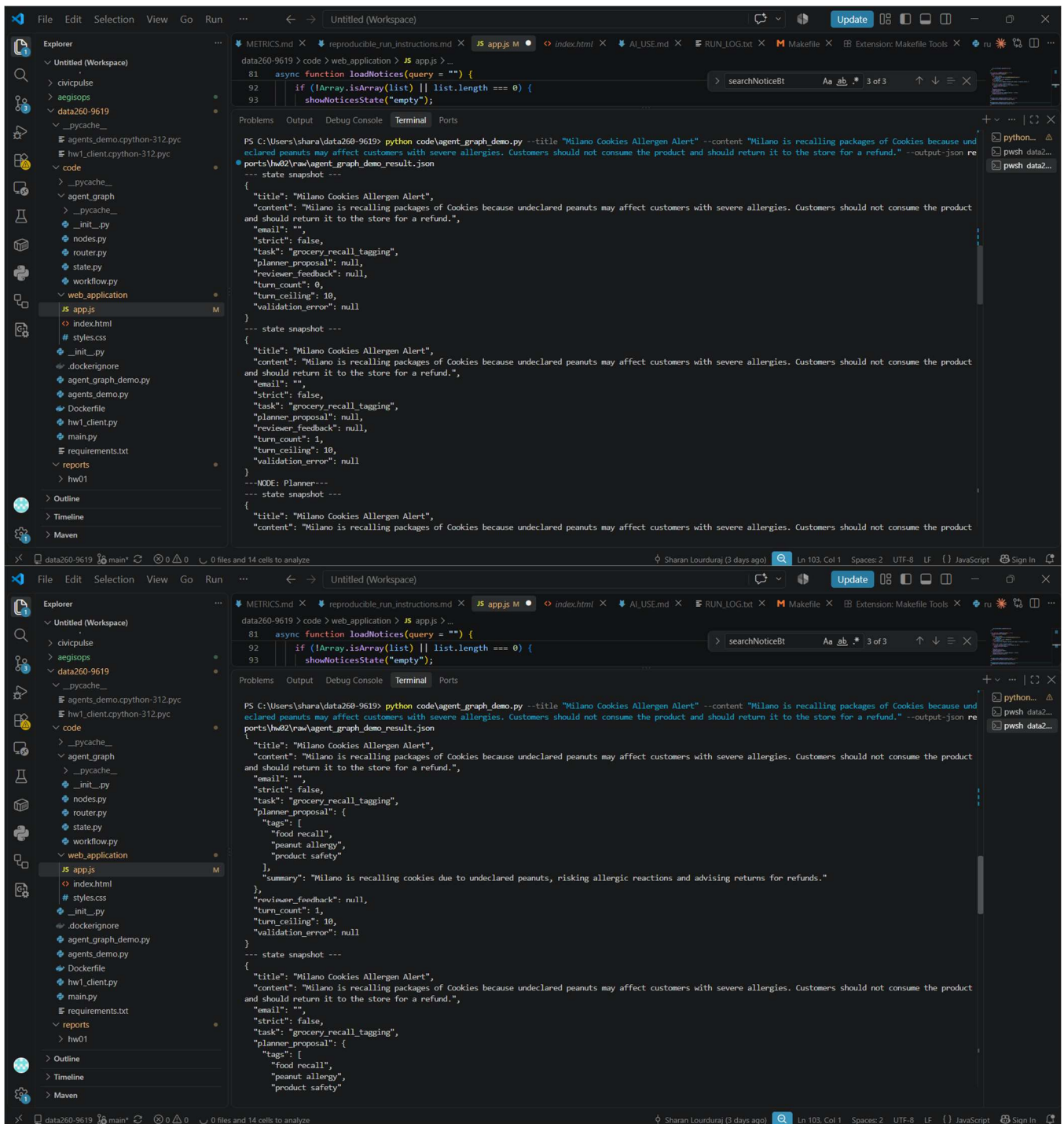
The graph has three nodes:

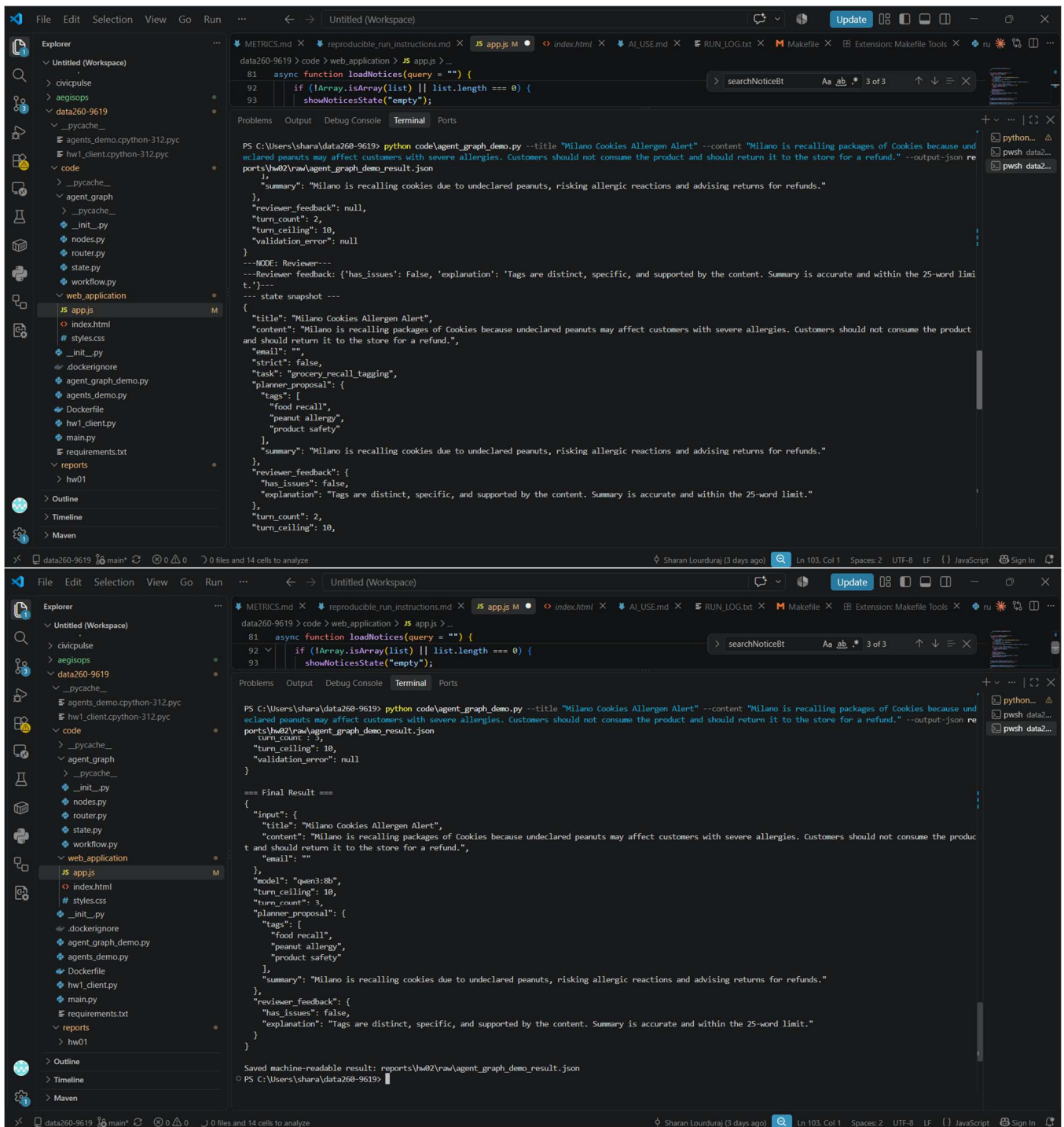
- Planner reads the title/content and proposes tags and a summary
- Reviewer checks whether the tags are distinct, grounded in the content, and whether the summary is accurate and within 25 words
- Supervisor a router that decides what happens next: send the state to the Planner, send it to the Reviewer, or stop

If the Reviewer finds a problem, the Supervisor sends the state back to the Planner for another attempt. A turn counter caps how many times this can happen (the turn_ceiling), so the graph cannot loop forever.

Normal run

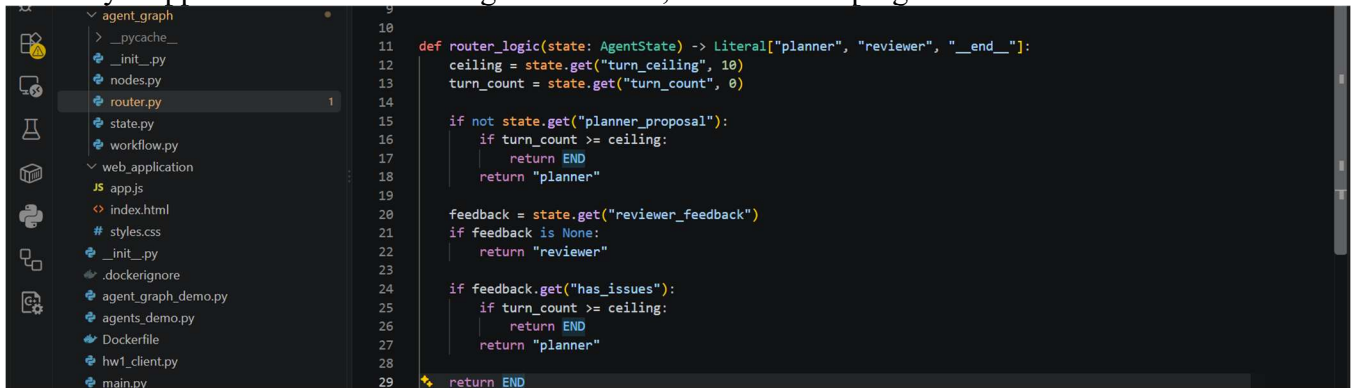
On a normal run, the Planner produces a valid proposal, the Reviewer approves it, and the graph ends. The console output below shows the full state at each step of the run.



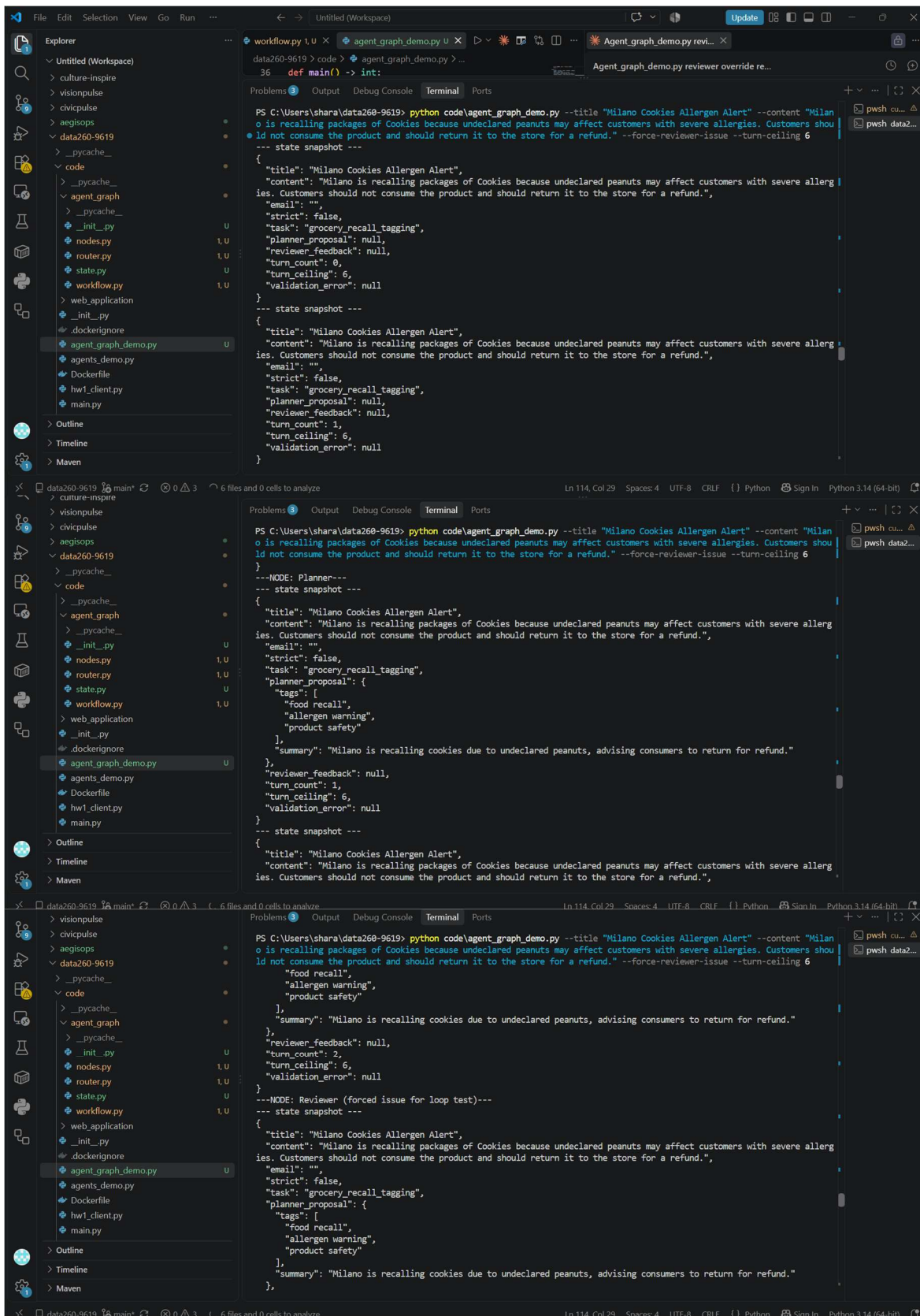


Self-correction loop (forced Reviewer rejection)

To prove the retry loop works, I ran the agent with a flag that forces the Reviewer to always report an issue. With --turn-ceiling 6, the graph looped Planner -> Reviewer -> Planner -> Reviewer twice, then correctly stopped once the turn ceiling was reached, instead of looping forever.

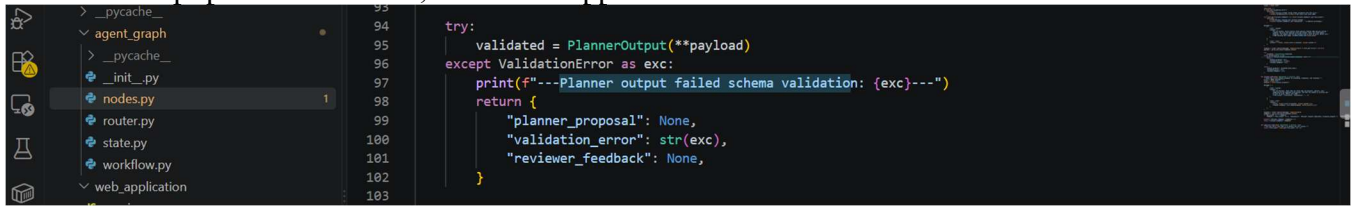


```
9
10
11 def router_logic(state: AgentState) -> Literal["planner", "reviewer", "__end__"]:
12     ceiling = state.get("turn_ceiling", 10)
13     turn_count = state.get("turn_count", 0)
14
15     if not state.get("planner_proposal"):
16         if turn_count >= ceiling:
17             return END
18         return "planner"
19
20     feedback = state.get("reviewer_feedback")
21     if feedback is None:
22         return "reviewer"
23
24     if feedback.get("has_issues"):
25         if turn_count >= ceiling:
26             return END
27         return "planner"
28
29     return END
```

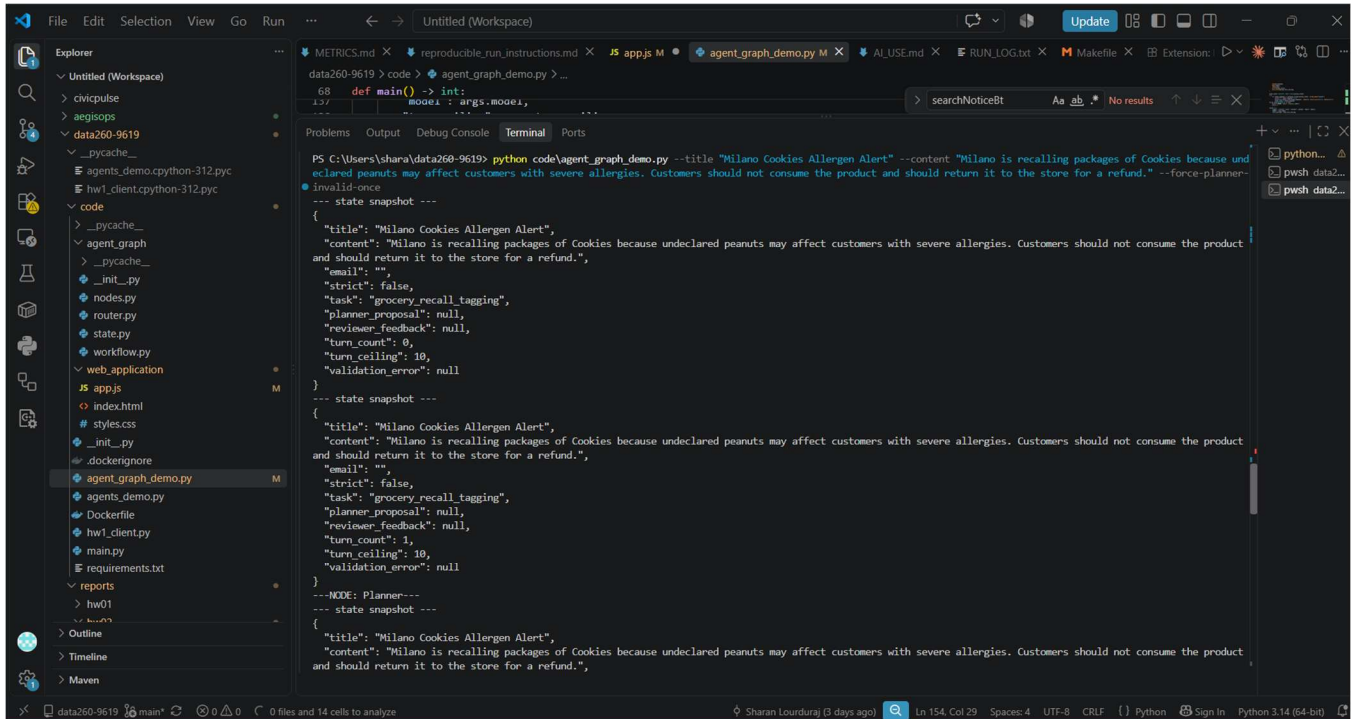


Schema-validation failure and retry

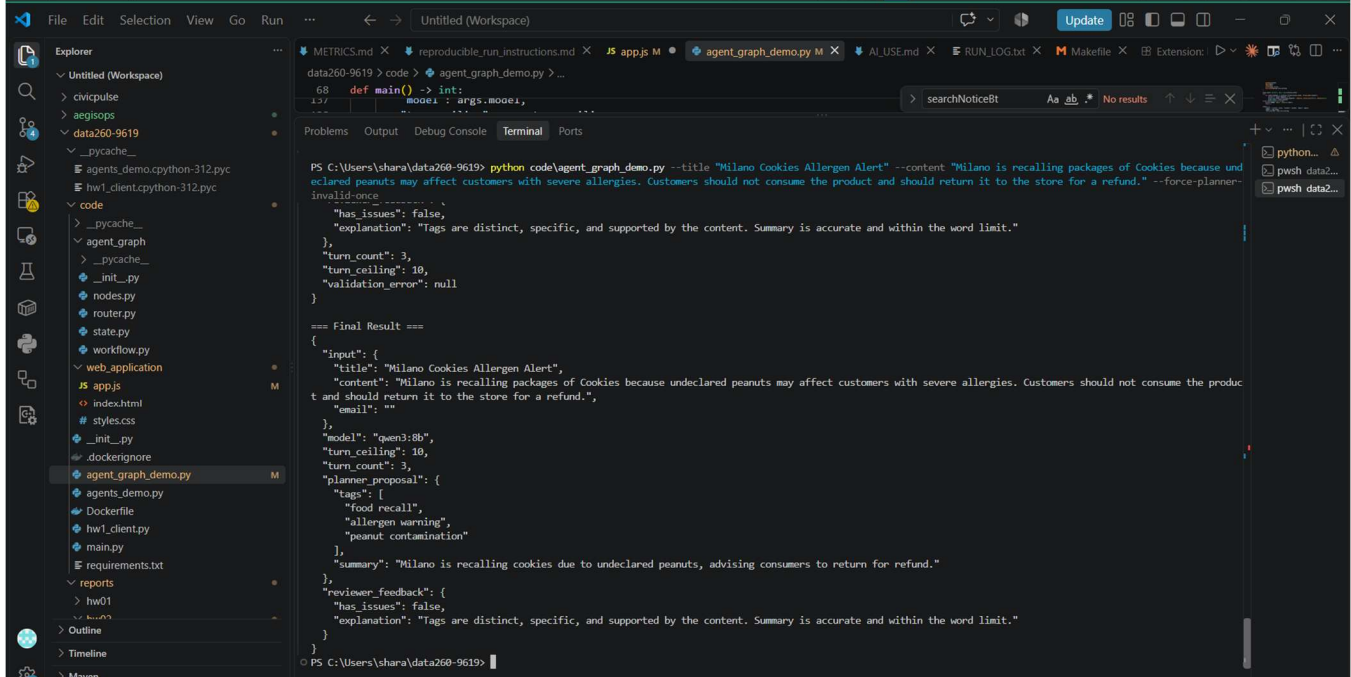
I also tested what happens when the Planner's own output fails schema validation (for example, if it returns only 2 tags instead of 3). I used a flag that corrupts the first Planner response on purpose so this would happen. The graph caught the real Pydantic validation error, fed it back to the Planner, and the second attempt produced a valid, Reviewer-approved result.



```
93
94
95     try:
96         validated = PlannerOutput(**payload)
97     except ValidationError as exc:
98         print(f'---Planner output failed schema validation: {exc}---')
99         return {
100             "planner_proposal": None,
101             "validation_error": str(exc),
102             "reviewer_feedback": None,
103         }
```



```
68 def main() -> int:
137     model = args.model,
138
139     --- state snapshot ---
140     {
141         "title": "Milano Cookies Allergen Alert",
142         "content": "Milano is recalling packages of Cookies because undeclared peanuts may affect customers with severe allergies. Customers should not consume the product and should return it to the store for a refund.",
143         "email": "",
144         "strict": false,
145         "task": "grocery_recall_tagging",
146         "planner_proposal": null,
147         "reviewer_feedback": null,
148         "turn_count": 0,
149         "turn_ceiling": 10,
150         "validation_error": null
151     }
152     --- state snapshot ---
153     {
154         "title": "Milano Cookies Allergen Alert",
155         "content": "Milano is recalling packages of Cookies because undeclared peanuts may affect customers with severe allergies. Customers should not consume the product and should return it to the store for a refund.",
156         "email": "",
157         "strict": false,
158         "task": "grocery_recall_tagging",
159         "planner_proposal": null,
160         "reviewer_feedback": null,
161         "turn_count": 1,
162         "turn_ceiling": 10,
163         "validation_error": null
164     }
165     ---NODE: Planner---
166     --- state snapshot ---
167     {
168         "title": "Milano Cookies Allergen Alert",
169         "content": "Milano is recalling packages of Cookies because undeclared peanuts may affect customers with severe allergies. Customers should not consume the product and should return it to the store for a refund.",
170         "email": "",
171         "strict": false,
172         "task": "grocery_recall_tagging",
173         "planner_proposal": null,
174         "reviewer_feedback": null,
175         "turn_count": 1,
176         "turn_ceiling": 10,
177         "validation_error": null
178     }
```



```
68 def main() -> int:
137     model = args.model,
138
139     --- state snapshot ---
140     {
141         "title": "Milano Cookies Allergen Alert",
142         "content": "Milano is recalling packages of Cookies because undeclared peanuts may affect customers with severe allergies. Customers should not consume the product and should return it to the store for a refund.",
143         "email": "",
144         "strict": false,
145         "task": "grocery_recall_tagging",
146         "planner_proposal": null,
147         "reviewer_feedback": null,
148         "turn_count": 0,
149         "turn_ceiling": 10,
150         "validation_error": null
151     }
152     --- state snapshot ---
153     {
154         "title": "Milano Cookies Allergen Alert",
155         "content": "Milano is recalling packages of Cookies because undeclared peanuts may affect customers with severe allergies. Customers should not consume the product and should return it to the store for a refund.",
156         "email": "",
157         "strict": false,
158         "task": "grocery_recall_tagging",
159         "planner_proposal": null,
160         "reviewer_feedback": null,
161         "turn_count": 1,
162         "turn_ceiling": 10,
163         "validation_error": null
164     }
165     ---NODE: Planner---
166     --- state snapshot ---
167     {
168         "title": "Milano Cookies Allergen Alert",
169         "content": "Milano is recalling packages of Cookies because undeclared peanuts may affect customers with severe allergies. Customers should not consume the product and should return it to the store for a refund.",
170         "email": "",
171         "strict": false,
172         "task": "grocery_recall_tagging",
173         "planner_proposal": null,
174         "reviewer_feedback": null,
175         "turn_count": 1,
176         "turn_ceiling": 10,
177         "validation_error": null
178     }
```

PART IV | Experiments

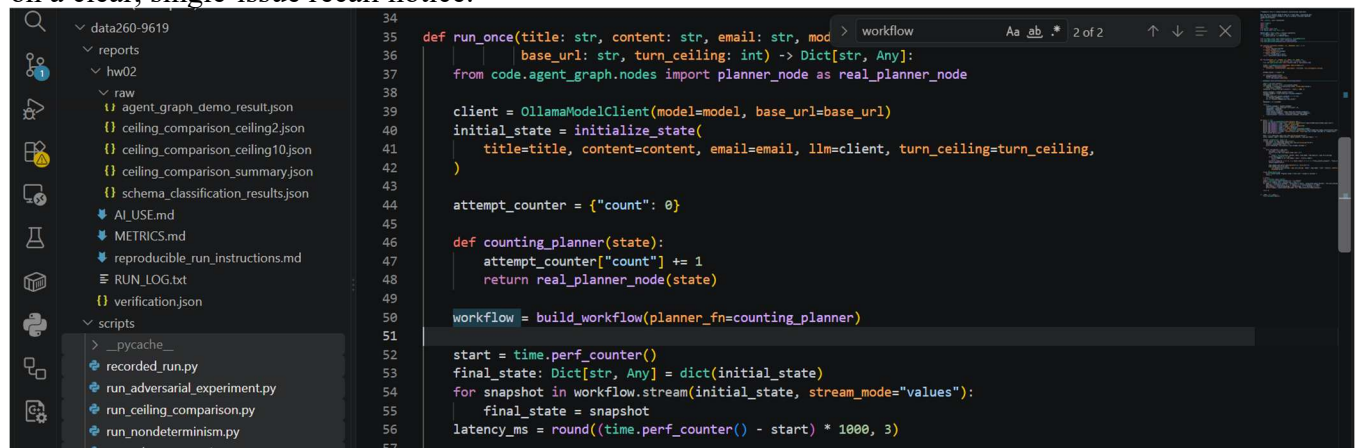
For this part I ran three experiments against the agent to see how reliable it is.

Experiment 1 Schema-validation classification (30 runs, same fixed input)

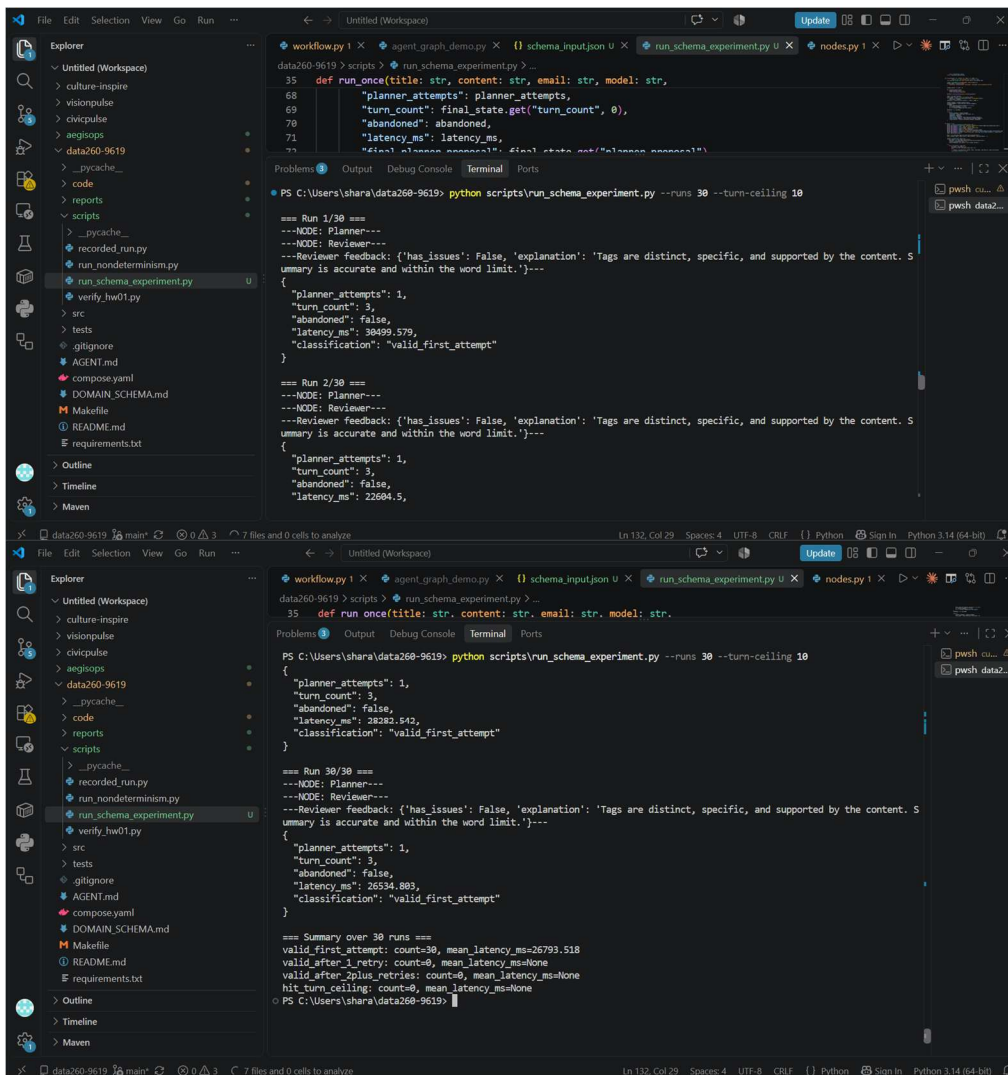
I used one fixed input (a Milano Cookies allergen-recall notice) and ran the full agent 30 times with a turn ceiling of 10, counting exactly how many times the Planner ran on each attempt.

Outcome over 30 runs	Count	Mean latency (ms)
Valid first attempt	30	26793.518
Valid after 1 retry	0	—
Valid after 2+ retries	0	—
Hit turn ceiling	0	—

On this fixed input, the local model produced a schema-valid, Reviewer-approved proposal on the first Planner attempt every single time. No retries were needed at all, which shows the model is very reliable on a clear, single-issue recall notice.



```
34 def run_once(title: str, content: str, email: str, model: str, base_url: str, turn_ceiling: int) -> Dict[str, Any]:
35     from code.agent_graph.nodes import planner_node as real_planner_node
36
37     client = OllamaModelClient(model=model, base_url=base_url)
38     initial_state = initialize_state(
39         title=title, content=content, email=email, llm=client, turn_ceiling=turn_ceiling,
40     )
41
42     attempt_counter = {"count": 0}
43
44     def counting_planner(state):
45         attempt_counter["count"] += 1
46         return real_planner_node(state)
47
48     workflow = build_workflow(planner_fn=counting_planner)
49
50     start = time.perf_counter()
51     final_state: Dict[str, Any] = dict(initial_state)
52     for snapshot in workflow.stream(initial_state, stream_mode="values"):
53         final_state = snapshot
54     latency_ms = round((time.perf_counter() - start) * 1000, 3)
```

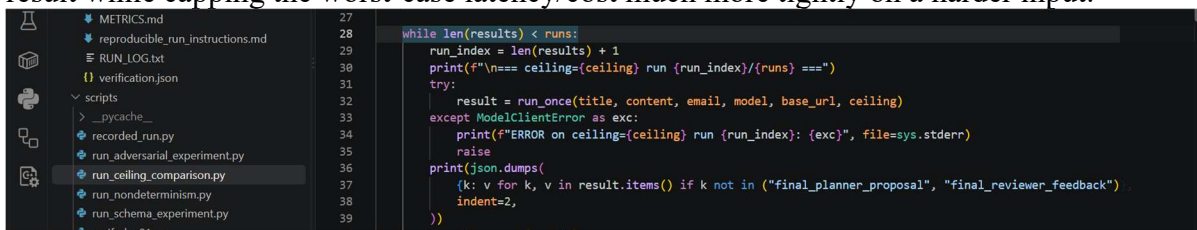


Experiment 2 Turn-ceiling comparison (ceiling 2 vs. ceiling 10, 20 runs each)

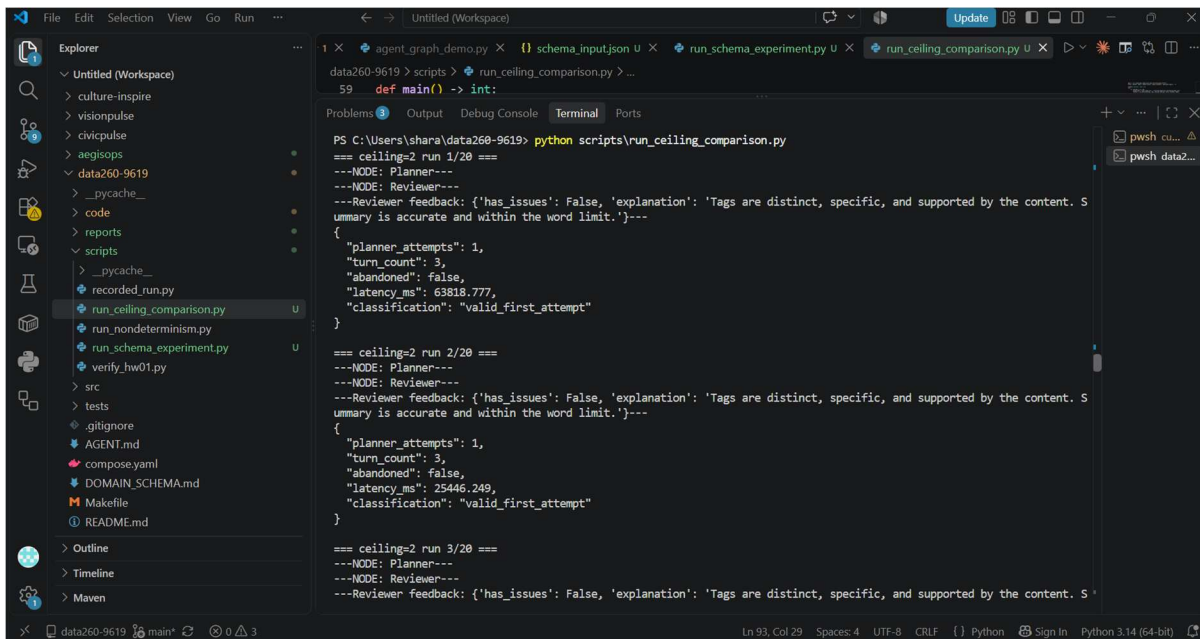
I ran the same fixed input 20 times with a turn ceiling of 2, and another 20 times with a turn ceiling of 10, to see whether a stricter ceiling would hurt reliability.

Turn ceiling	Completion rate	Mean latency (ms)	n
2	1.00	28626.236	20
10	1.00	26373.833	20

Both ceilings reached 100% completion with basically the same latency (the small difference is normal run-to-run variance, not retries none happened at either ceiling on this input). Since raising the ceiling from 2 to 10 didn't improve reliability at all, I would deploy with a turn ceiling of 2 it gives the same result while capping the worst-case latency/cost much more tightly on a harder input.

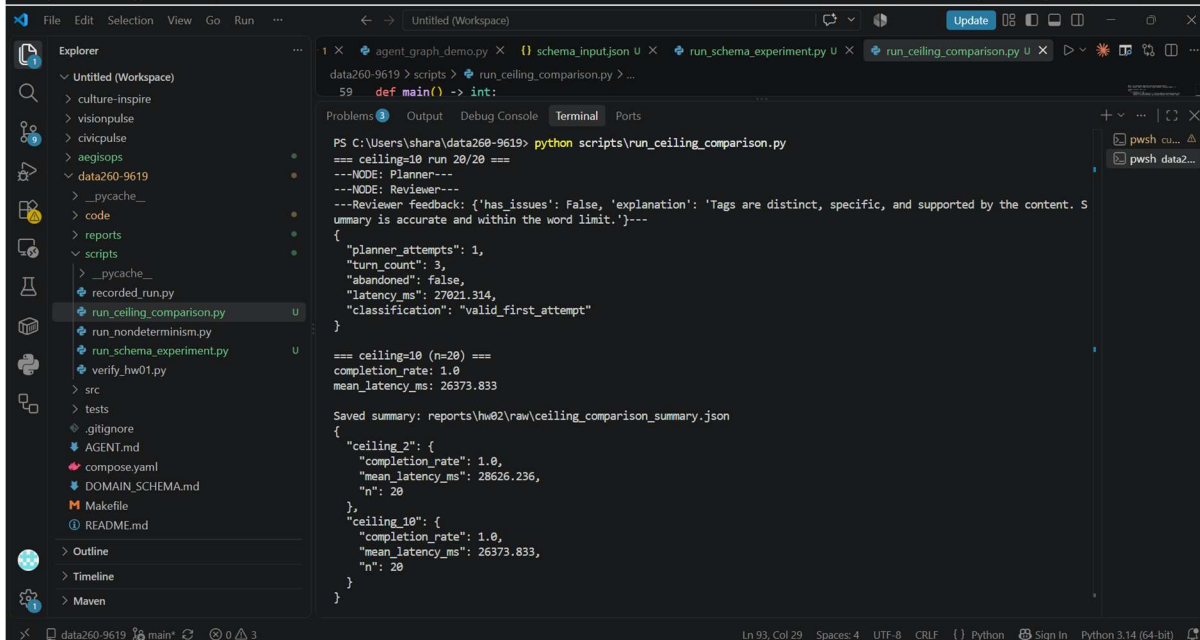


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```



```
data260-9619 > scripts > run_ceiling_comparison.py > ...
59 def main() -> int:

Problems Output Debug Console Terminal Ports
PS C:\Users\shara\data260-9619> python scripts/run_ceiling_comparison.py
=== ceiling=2 run 1/20 ===
---NODE: Planner---
---NODE: Reviewer---
---Reviewer feedback: {'has_issues': False, 'explanation': 'Tags are distinct, specific, and supported by the content. Summary is accurate and within the word limit.'}---
{
  "planner_attempts": 1,
  "turn_count": 3,
  "abandoned": false,
  "latency_ms": 63818.777,
  "classification": "valid_first_attempt"
}
=== ceiling=2 run 2/20 ===
---NODE: Planner---
---NODE: Reviewer---
---Reviewer feedback: {'has_issues': False, 'explanation': 'Tags are distinct, specific, and supported by the content. Summary is accurate and within the word limit.'}---
{
  "planner_attempts": 1,
  "turn_count": 3,
  "abandoned": false,
  "latency_ms": 25446.249,
  "classification": "valid_first_attempt"
}
=== ceiling=2 run 3/20 ===
---NODE: Planner---
---NODE: Reviewer---
---Reviewer feedback: {'has_issues': False, 'explanation': 'Tags are distinct, specific, and supported by the content. Summary is accurate and within the word limit.'}---
```



```
data260-9619 > scripts > run_ceiling_comparison.py > ...
59 def main() -> int:

Problems Output Debug Console Terminal Ports
PS C:\Users\shara\data260-9619> python scripts/run_ceiling_comparison.py
=== ceiling=10 run 20/20 ===
---NODE: Planner---
---NODE: Reviewer---
---Reviewer feedback: {'has_issues': False, 'explanation': 'Tags are distinct, specific, and supported by the content. Summary is accurate and within the word limit.'}---
{
  "planner_attempts": 1,
  "turn_count": 3,
  "abandoned": false,
  "latency_ms": 27021.314,
  "classification": "valid_first_attempt"
}
=== ceiling=10 (n=20) ===
completion_rate: 1.0
mean_latency_ms: 26373.833

Saved summary: reports\hw02\raw\ceiling_comparison_summary.json
{
  "ceiling 2": {
    "completion_rate": 1.0,
    "mean_latency_ms": 28626.236,
    "n": 20
  },
  "ceiling 10": {
    "completion_rate": 1.0,
    "mean_latency_ms": 26373.833,
    "n": 20
  }
}
```

Experiment 3 Adversarial input (5 runs, turn ceiling 2)

For this one I intentionally designed a hard input: a single notice that bundles four unrelated incidents together (a *Listeria* contamination, an undeclared-allergen issue, an unconfirmed carbonation anomaly, and a labelling mix-up, across four different facilities).

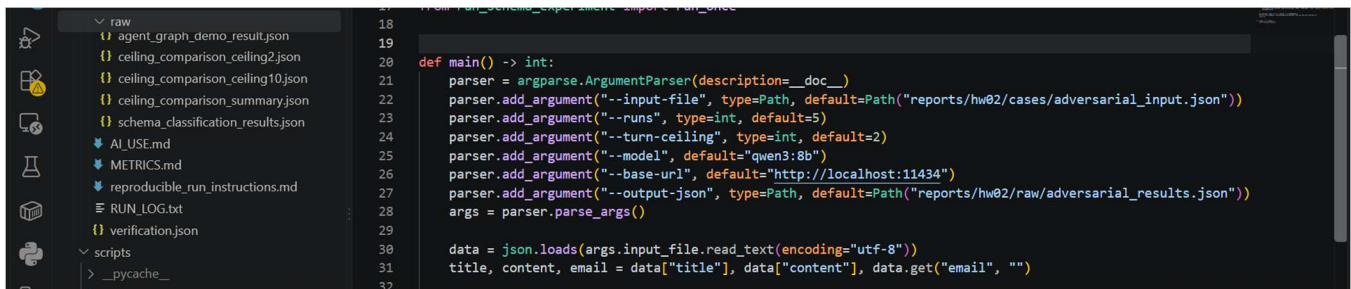
Result: 5 out of 5 runs hit the turn ceiling. Every run's proposal was rejected by the Reviewer on the first attempt, and with a ceiling of 2 there was no room left for a second try.

Some of the actual Reviewer rejection reasons from the runs:

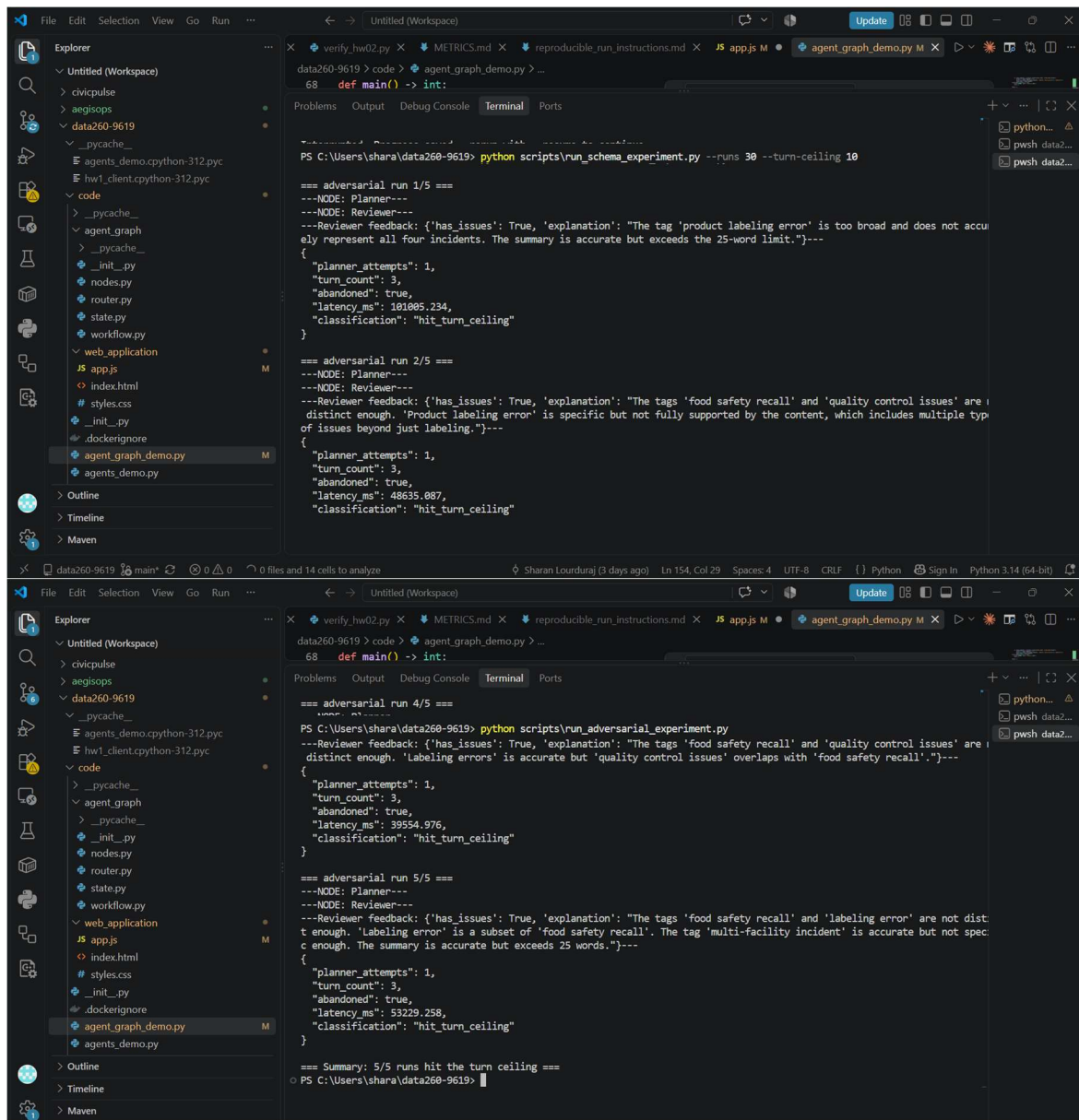
- “The tags 'product recall' and 'labelling error' are not distinct enough and the 'food safety incident' tag is too generic. The summary is accurate but exceeds 25 words.”
- “The tags 'food safety recall' and 'quality control issues' are not distinct enough... 'quality control issues' overlaps with 'food safety recall'.”
- “'Labeling Errors' is a subset of 'Quality Control Issues', and the summary is inaccurate and exceeds 25 words.”

Why this happens: the notice crams four unrelated incidents into one notice, but the schema only allows exactly 3 tags and a 25-word summary. The Planner can't satisfy both at once general tags that cover all four incidents end up overlapping and “not distinct enough,” naming all four incidents blows past the word limit, and a tag grounded in only one incident gets flagged as incomplete. This is a structural mismatch between the fixed output shape and how complex the input is, not random bad luck all 5 runs failed for the same reason.

A higher turn ceiling would not fix this, since the problem isn't a one-off mistake that a retry can correct. A better fix would be to detect when a notice bundles multiple unrelated incidents and run each incident through its own single-issue tagging pass, instead of forcing one Planner call to compress everything into 3 tags and 25 words.

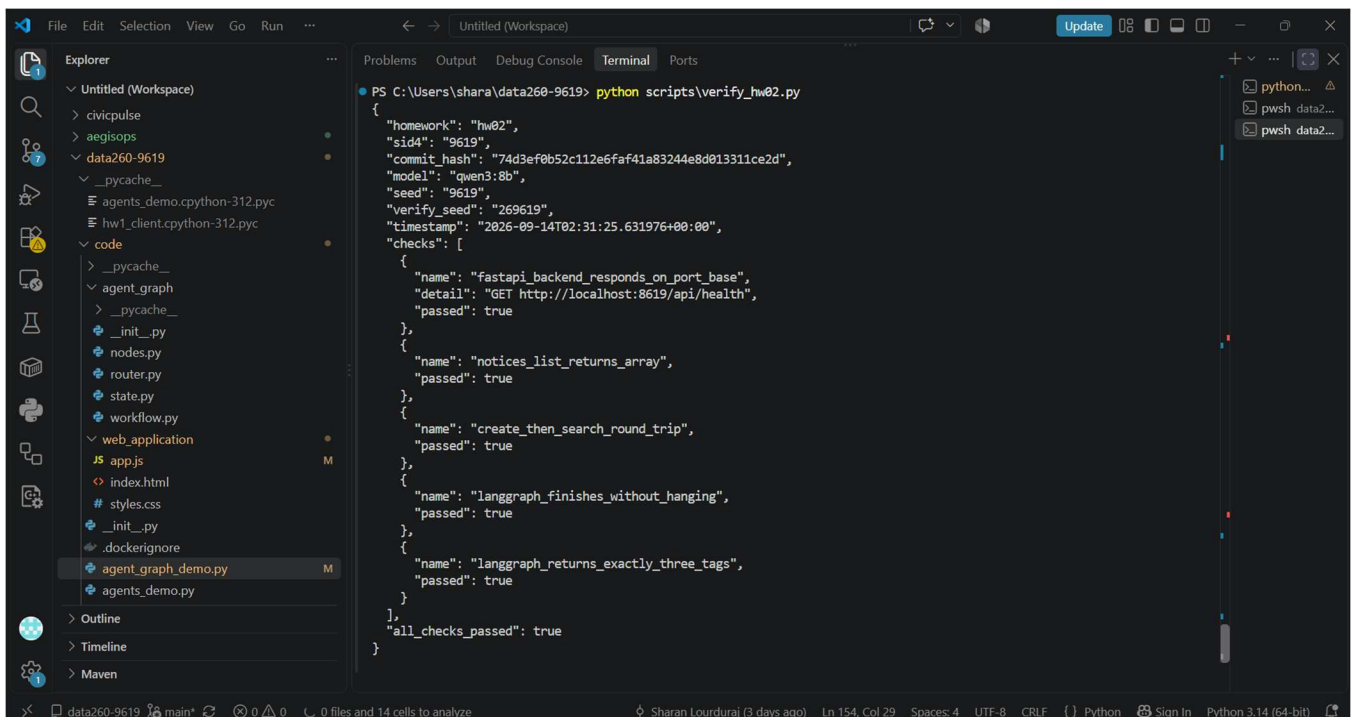


```
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22 def main() -> int:
23     parser = argparse.ArgumentParser(description=__doc__)
24     parser.add_argument("--input-file", type=Path, default=Path("reports/hw02/cases/adversarial_input.json"))
25     parser.add_argument("--runs", type=int, default=5)
26     parser.add_argument("--turn-ceiling", type=int, default=2)
27     parser.add_argument("--model", default="qwen3:8b")
28     parser.add_argument("--base-url", default="http://localhost:11434")
29     parser.add_argument("--output-json", type=Path, default=Path("reports/hw02/raw/adversarial_results.json"))
30     args = parser.parse_args()
31
32     data = json.loads(args.input_file.read_text(encoding="utf-8"))
33     title, content, email = data["title"], data["content"], data.get("email", "")
```



Verification

Before finishing, I ran `scripts/verify_hw02.py`, which starts the backend, checks the notices API, confirms a create-then-search round trip works, and confirms the LangGraph agent finishes without hanging and returns exactly 3 tags. All checks passed.



The screenshot shows a VS Code editor window with a workspace named "Untitled (Workspace)". The Explorer panel on the left shows a file tree with folders like "civicpulse", "aegisops", and "data260-9619". The "code" folder is expanded, showing files like "agent_graph_demo.py" and "agents_demo.py". The main editor area displays the output of a Python script, which is a JSON object containing a list of checks and their results. The checks are: "fastapi_backend_responds_on_port_base", "notices_list_returns_array", "create_then_search_round_trip", "langgraph_finiishes_without_hanging", and "langgraph_returns_exactly_three_tags". All checks passed.

```
PS C:\Users\shara\data260-9619> python scripts/verify_hw02.py
{
  "homework": "hw02",
  "sid4": "9619",
  "commit_hash": "74d3ef0b52c112e6faf41a83244e8d013311ce2d",
  "model1": "qwen3:8b",
  "seed": "9619",
  "verify_seed": "269619",
  "timestamp": "2026-09-14T02:31:25.631976+00:00",
  "checks": [
    {
      "name": "fastapi_backend_responds_on_port_base",
      "detail": "GET http://localhost:8619/api/health",
      "passed": true
    },
    {
      "name": "notices_list_returns_array",
      "passed": true
    },
    {
      "name": "create_then_search_round_trip",
      "passed": true
    },
    {
      "name": "langgraph_finiishes_without_hanging",
      "passed": true
    },
    {
      "name": "langgraph_returns_exactly_three_tags",
      "passed": true
    }
  ],
  "all_checks_passed": true
}
```